Abatement coefficient g in RCC-MRx

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[1]**Octave**
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Abstract

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Don't forget to mention the nuclear context, which is the central topic around which all publications on this issue of the journal focus on

Keywords: high temperature piping analysis ; RCC-MRx; elastic follow-up; abatement coefficient ; plasticity.

1 Introduction

Steel pipings are metallic structures (usually pressurized) consisting of straight tubes with hollow cicular cross-sections, assembled by means of welds, fittings (elbows, reducers, tees, etc.) or flanged connections. Due to functional and safety requirements, components such as valves, flow meters, expansion joints and other in-line components might be assembled on piping tubes. The entire structure can be connected to pressurized equipments by means of nozzles and supported by dedicated support structures. Together, these elements constitute a large and complex structure that must be addressed through rigorous engineering analysis.

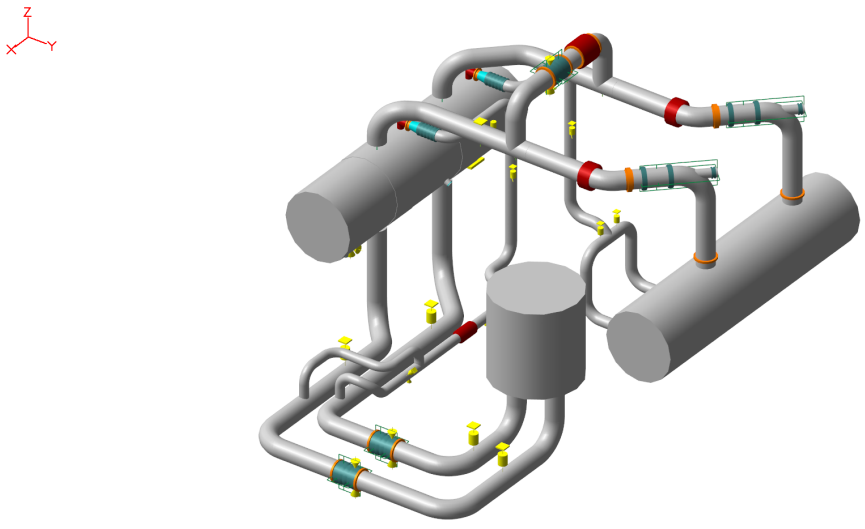


Figure 1: A typical piping structure modelled under PEPS 8.0

1.1 Calculation methodology in standard piping analysis

Piping stress engineers are interested in the stress state at every material point within pipings that might extend over kilometers. Performing a detailed 3D finite element analysis (FEA) of such structures would be excessively time-consuming and is generally incompatible with the industrial projects' needs. In this context, piping design

codes provide rules for the analysis of piping systems, generally based on the assumption of **linear elastic material behavior** (Hooke's law), with allowable stress limits established to prevent the various failure modes and to ensure an adequate level of safety.

Based on the nature of the loading, piping stress analysis distinguishes between *primary* loads (pressure, weight, seismic inertia, etc.) and *secondary* loads (such as thermal expansion). Primary stresses balance the mechanical loads to get equilibrium. Secondary stresses are known for their self-limiting characteristic.
Figure 2 shows an example of the two stresses. **E** presents the stress state of a material point predicted by an elastic approach. **P** and **S** are stress states responding respectively to primary and secondary loads, considering the elastoplastic law.
Primary loads are force-imposed in nature and are balanced by a stress field $\sigma_P = \sigma_E$, regardless of whether the material behavior is described by an elastic or an elastoplastic constitutive law. However, the corresponding elastoplastic strain ε_P might be significantly larger than the elastically predicted strain ε_E, under high primary loading conditions, due to plastic flow. As a result, **excessive deformation** may develop at the structural scale when local plastified zones expand. If the load continues to increase, the stable execessive deformation regime might be lost as structural softenings progressively dominate the global response, eventually lead the structure into the **plasticity instability** damage mode. Moreover, primary stresses are also responsible for **buckling** (also known as elastic or elastoplastic instability), due to the presence of unrexable stresses with geometry instabilities.

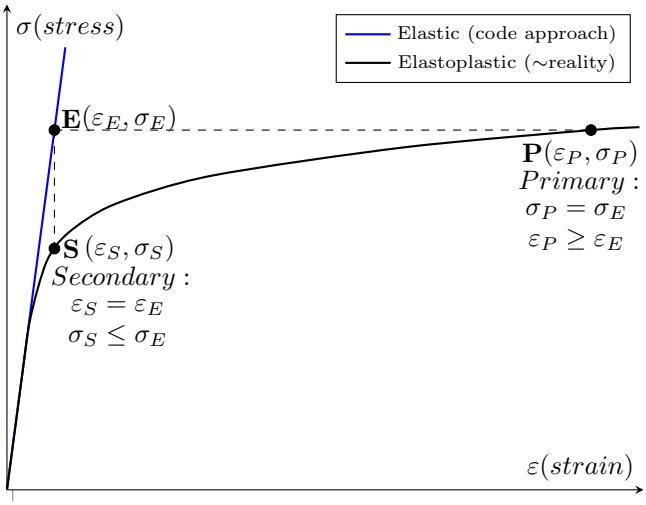


Figure 2: Stress categorization: primary and secondary stresses

Secondary loads are displacement-controlled by nature, meaning that the structure is subjected to strain-imposed boundary conditions. As a result, the structure deformation is prescribed and $\varepsilon_S = \varepsilon_E$ regardless of the type of analysis employed. Under sufficiently high secondary loads, localized yielding may occur and plastic flow can develop, leading to elastoplastic stress levels lower than those predicted by a purely elastic analysis: $\sigma_S \leq \sigma_E$. The remainder of the structure, constrained by the imposed-strain boundary conditions, accommodates the local plastic deformation and limits the extent of the plastified region. Secondary stresses are threfore **self-limiting**. Secondary stresses are usually induced by (cyclic) thermal expansion, seismic anchor movements, or differential settlement. The two most significant damage modes are **fatigue** under cyclic loading and **progressive deformation**.
Piping design codes classify loads according to these two categories and specify allowable stress limits based on the associated failure modes, drawing on both experimental evidence and theoretical considerations. In practice, a linear elastic analysis is used to determine the stress state at each point of the structure (**E**). The nature of the loading can then be identified, allowing the actual response to be interpreted according to the characteristics of primary (**P**) or secondary stresses (**S**) and assessed against the corresponding code compliance criteria. This methodology helps avoid the need for complex nonlinear elastoplastic analyses.

The previous paragraphs briefly outlined the main ideas behind piping design codes for addressing the complex structural problems encountered in piping analysis, in particular the stress categorization within a linear approach. A further challenge arises in numerical simulation: even linear 3D finite element analyses (FEA) remain computationally expensive, in terms of model preparation, solution time, and result post-processing. In this context, beam theory plays a key role, providing an efficient means of representing long pipe runs

and their associated components while significantly reducing the computational effort required. The classical Euler–Bernoulli hypothesis neglects transverse shear deformation and is only valid for slender members. The Timoshenko beam theory retains transverse shear and remains valid for stockier members as well. Standard nuclear piping codes rely on Timoshenko-type beam elements for the straight and curved segments of the model. Beam theory was originally developed for straight pipes and must be adapted for curved pipes (bend or elbows). Under bendings, elbows undergoes cross-sectional ***ovalisation***: the initial circular cross-section distorts into an oval shape. This ovalisation is a geometrical non-linearity, illustrated by the shell FEA analysis shown in Figure 3, increases the flexibility of the elbow well beyond the prediction of classical (straight) beam theory. The ***flexibility factor*** k is defined as the ratio between the rotation per unit length of the curved component and the rotation per unit length of an equivalent straight pipe under the same bending moment. Von Kármán/Hovgaard suggested first approximations to estimate the flexibility factor of elbows, and latter on Beskin introduced simpler reference approximation. In the slide sequence, the latter is presented as the practical formula retained for piping calculations, without an explicit intermediate derivation from the former. The flexibility factor of a smooth, unpressurised elbow is therefore written in the benchmark, in terms of the non-dimensional flexibility characteristic h, as

$$h = \frac{t\,R}{r^2}, \qquad k = \frac{1.65}{h}, \tag{1}$$

where k is the bend flexibility factor, t is the wall thickness, r the mean cross-sectional radius, and R the bend radius. The flexibility factor k will be considered in the stiffness matrix of the Timoshenko beam model. Finally,

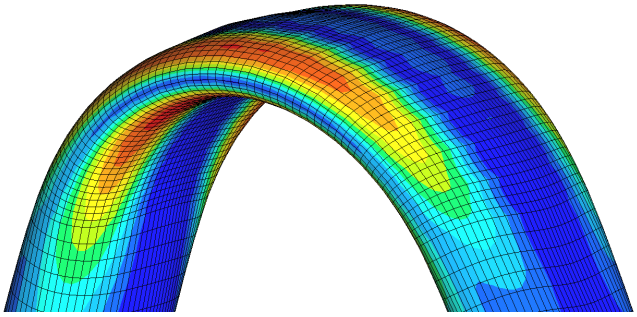


Figure 3: Finite-element illustration of elbow cross-section ovalisation under in-plane bending: the stress concentration follows the ovalised region around the crown of the bend.

because of the geometric and material discontinuities near fittings, welds and bends, the local stress in these areas is amplified with respect to the nominal beam-theory stress. This amplification is expressed through the *Stress Intensification Factor* (SIF), i. Following Markl's theoretical works and experimental tests.

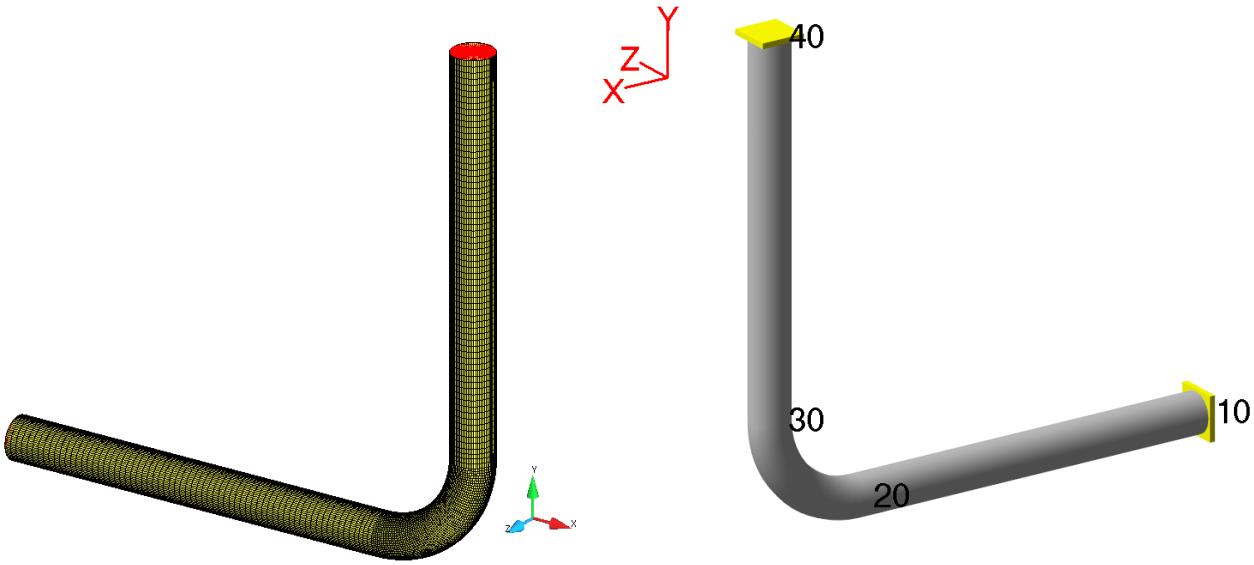
Present the impact of pressure on the ovalization here + introduce the pressure ovalization correction

Note: it might be impossible to consider pressure in this work. Because Marc will have to explicitly consider pressure load, but here we only want the pressure effects on the elbows' flexibility (deformation) and consider only thermal exp –> to discuss more

1.2 Elastic benchmark

Maybe say that we are referring to the anlaysis of point E in Figure 2

The objective of the benchmark presented below is is to demonstrate the consistency between a Timoshenko beam model (Figure 4a) and a 3D shell FEA model (Figure 4b), for the analysis of a simple stainless-steel piping structure.



(a) FEA shell model - MSC Marc

(b) FEA beam model - PEPS 8.0

Figure 4: Geometry and finite element model.

The configuration consists of two straight pipe segments (10-20 and 30-40) connected by an elbow (20-30). The structure is made of A3.3S material (RCC-MRx Ed. 2022), with an outer diameter of 219.07 mm, a wall thickness of 3.76 mm, a straight-pipe length of 2000 mm for each segment, and an elbow radius of 500 mm. The straight-pipe ends at nodes 10 and 40 are fully anchored. The piping is uniformly heated from an ambient temperature of 20°C to an in-service temperature of 550°C. Only thermal expansion is considered in this analysis, and the material behavior is assumed to be linear elastic.

Two independent numerical models are used and cross-checked:

- **Aspect Nuclear Pipe Stress** ("NS" solver, PEPS 8.0), a Timoshenko-beam FEA piping model in which the elbow flexibility is represented through the analytical flexibility factor k of Eq. (1);
- **MSC Marc**, a shell finite-element model using bilinear thick-shell elements (four-node, Marc Element 75), in which elbow ovalisation is captured directly by the shell discretisation rather than through an analytical flexibility factor. This element uses bilinear interpolation of coordinates, displacements and rotations, with membrane strains from the displacement field and curvatures from the rotation field; transverse shear strains are evaluated at the mid-edges and interpolated to the Gauss integration points, giving an efficient element that remains accurate in the thin-shell limit while avoiding tying constraints at plate intersections. The two ends of the model are connected to the FE mesh through RBE2 rigid-body elements, as shown in Figure **??**. RBE2 links can locally generate spurious over-stiffening and stress concentrations; this is considered acceptable here because stresses are not examined at the anchor points themselves (only the elbow region is of interest for stress results), so that the simpler RBE2 connection is preferred over an RBE3 connection.

Before introducing plasticity, the two models are calibrated against one another on a purely elastic basis. Because the beam model represents elbow ovalisation only through the analytical flexibility factor k, whereas the shell model captures it directly through the finite-element discretisation, the anchor reaction moment (at the RBE2 master node) is used as the key comparison quantity, since it is highly sensitive to the overall flexibility of the system.

Table 1 reports the anchor reaction moment obtained with the beam model with and without the elbow flexibility factor, and with the shell model at increasing circumferential mesh densities.

Table 1: Anchor reaction moment: beam model (with/without flexibility factor) versus shell FE model at increasing mesh density.

Model	**Mz [N m]**	**FX [N]**	**FY [N]**
PEPS beam model, without k-factor	56 702	xxx	yyy
PEPS beam model, with k-factor	39 632	xxx	yyy
Marc shell, 80 elements/circumference	40 290	xxx	yyy

The presentation also documents the same run carried through to its final increment on the finer shell mesh, reproduced in Figure **??**: the maximum displacement stabilises at 35.25 mm (node 8121), essentially unchanged from the coarser/earlier-increment view of Figure **??**, confirming that the elastic solution is converged with respect to load increment.

The beam model without the elbow k-factor overestimates the system stiffness by about 40 % relative to the converged shell solution, confirming the essential role of elbow ovalisation in the overall flexibility of the system. Once the analytical flexibility factor of Eq. (1) is introduced ($k = 10.1717$), the beam-model reaction moment (39 632 N m) matches the shell result at a moderate mesh density (20 elements per circumference, 39 650 N m) extremely well, confirming both the validity of the analytical flexibility factor and the adequacy of the beam-element representation for this geometry. Figure **??** shows the corresponding shell-model stress field and displacement pattern. As the shell mesh is further refined to 40 and 80 elements per circumference, the reaction moment increases moderately further (to 40 180 N m and 40 290 N m, respectively), indicating a degree of mesh sensitivity in the shell model that is revisited in Section **??**. Large-deformation effects (ovalisation) are included in the shell analysis.

The presentation also compares the global deformed shapes produced by the two models. The beam-based NS model predicts a maximum thermal-expansion displacement of 33.649 mm at point Z010, reported in Figure 5, which is consistent with the shell-model maximum displacement reported in Figure **??**; this provides a second, kinematic confirmation that the calibrated beam model reproduces the shell-model flexibility with acceptable accuracy for this benchmark. Figure 22 further documents the shell mesh that was finally retained for the non-linear run.

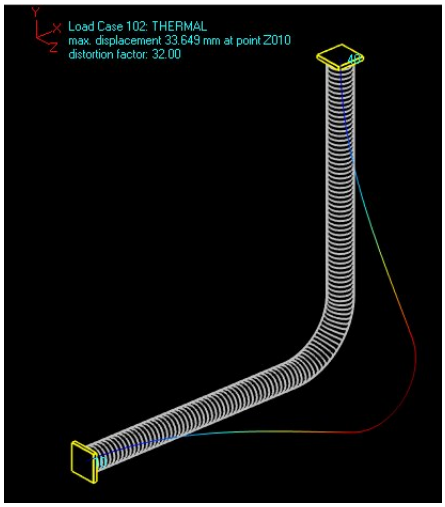


Figure 5: Deformed shape from the Aspect Nuclear Pipe Stress elastic benchmark (load case 102, thermal expansion): the beam-model maximum displacement is 33.649 mm at point Z010, close to the shell-model prediction after calibration of the elbow flexibility.

Show more results here
Conclusion of the elastic benchmark here
Show that we are looking for point E solution, with a much less time-consuming approach (PEPS 8.0)

2 Elastic follow-up

Section 1 briefly summarized the approach adopted by most standard piping codes for the assessment of piping stresses. For piping systems used in nuclear power plants, more stringent requirements apply, requiring more sophisticated calculation methodologies, particularly for high-temperature applications. One of the main differences is the need to consider the so-called **elastic follow-up**.

In piping systems, geometrical discontinuities or load-induced deformations (e.g., elbow ovalization) can lead to stress concentrations at specific locations, such as elbows, tees, reducers, or thickness transitions. Under sufficiently high loading, most of the structure may remain in the elastic regime and behave as linear springs, thereby elastically following up and constraining these highly stressed regions where local plastic deformation develops. This phenomenon is referred to as elastic follow-up [8] [9] .

In this section, the elastic follow-up concept will be presented based on the work of Roland L. Roche [4], without considering the impacts of creep and stress relaxation.

The Ramberg–Osgood constitutive law:

to redo this paragraph on Ramberg-Osgood

To model the elasto-plastic material response, the Ramberg–Osgood law is used in its general textbook form, decomposing the total strain into an elastic and a power-law plastic part,

$$\varepsilon_t = \varepsilon_e + \varepsilon_p = \frac{\sigma}{E} + \left(\frac{\sigma}{K}\right)^{1/n}, \tag{2}$$

with K a strength coefficient and n a hardening exponent. The source presentation specialises Eq. (2) to RCC-MRx's own parametrisation of the strength coefficient as a temperature-dependent multiple of the 0.2% proof stress,

$$K(T) = C_0\, R_{p0.2}(T), \tag{3}$$

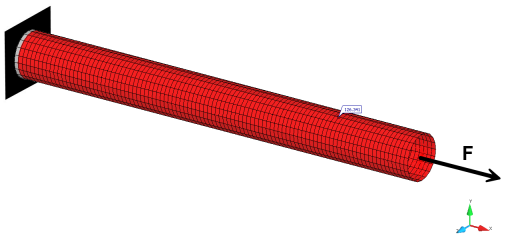
and to the code's own definition of the plastic-strain scale (the leading 0.01, or 100×, factor below reflects the code's percent-strain convention), giving the exact form used in the benchmark computations:

$$\varepsilon(\sigma) = \frac{\sigma}{E} + 0.01\left(\frac{\sigma}{C_0\, R_{p0.2,\mathrm{mean}}}\right)^{1/n_0}, \tag{4}$$

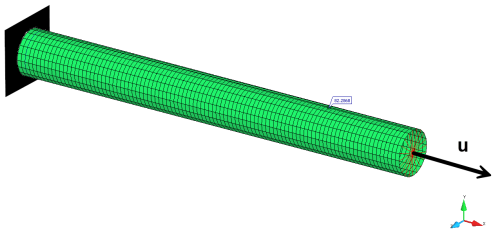
with, for 316L(N) stainless steel, $n_0 = 0.1125$ ($1/n_0 = 8.89$) and $C_0 = 1.198$; at the benchmark temperature of 550 °C, $R_{p0.2} \approx 106$ MPa, so that the stress scale $C_0 R_{p0.2}$ used in the actual benchmark computations of Eq. (4) is about 127 MPa. The temperature dependence therefore enters through $R_{p0.2}(T)$ in Eq. (3), whereas C_0 and n_0 retain the same normalised curve shape at all temperatures — i.e., no change of hardening *shape* with temperature is assumed. A separate sensitivity study reported in the presentation, evaluated at 500 °C (where $R_{p0.2} \approx 110$ MPa, so that the reference parameter set gives $C_0 R_{p0.2} \approx 131.8$ MPa), compares this reference curve with an alternative fit ($C_0 = 1.233$, $n_0 = 0.130$, i.e. $1/n_0 = 7.69$, $C_0 R_{p0.2} \approx 135.6$ MPa) calibrated on the same 500 °C tensile data; the resulting stress–strain curve is only marginally softer than the reference one over the whole strain range plotted, confirming that the choice of hardening parameters is not the dominant source of discrepancy discussed in Section **??**. –> to use less number/data to talk about Ramberg Osgood

2.1 Elastic-follow-up – deviation of the secondary stress category

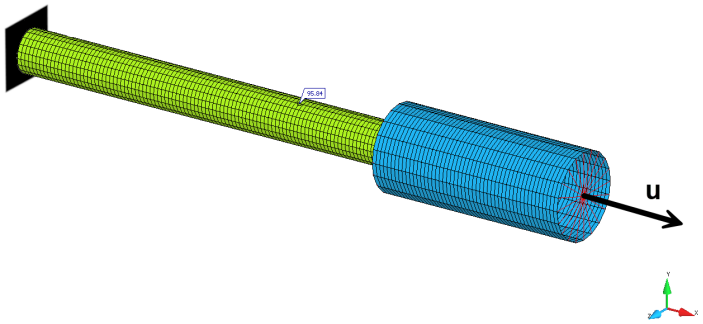
In order to illustrate elastic follow-up, Roland L. Roche [4] proposed a test involving three bars of identical geometry. Bar 1 is subjected to a primary load, while Bar 2 is subjected to a secondary load. Bar 3 is identical to the other two bars but is connected to an additional spring and is also subjected to a secondary load. In his conclusions, Roland L. Roche showed that Bar 3, although subjected only to a secondary load, exhibits an intermediate behavior between those associated with purely primary and purely secondary loading. With our own selected beam geometry and material, the test is reproduced and shown in figure 6.



(a) Beam 1 under primary load (force imposed F=320 kN) - purely primary stress response



(b) Beam2 under secondary load (displacement imposed u=1.623mm) - purely secondary stress response



(c) Beam3 under secondary load (displacement imposed u=8.115mm) - intermediate primary-secondary stress response

Figure 6: The three beams share a common geometry: tubes of 2 m length, 219.07 mm outside diameter, and 3.76 mm wall thickness. The material is A3.3S (RCC-MRx Edition 2022). Beam 3 includes an additional tubular section representing the spring, with a length of 1 m and an outside diameter of 434.38 mm; the wall thickness and material are identical to those of the other beams. The spring section remains purel elastic, whereas plasticity may develop in the common section. All beams are fixed at one end and loaded at the other.

The applied forces and imposed displacements for the three beams were selected such that, under a purely elastic analysis, all three beams exhibit the same stress and strain state **E**: $\varepsilon_E = 0.081\%$ and $\sigma_E = 126\,\text{MPa}$.

When the Ramberg–Osgood plasticity law is introduced into the common section, beam 1 (point **P**) exhibits a purely primary stress response, as expected: $\varepsilon_P = 1\% > \varepsilon_E$ and $\sigma_P = 126\,\text{MPa} = \sigma_E$. Similarly, beam 2 (pont **S**) exhibits a purely secondary stress response: $\varepsilon_S = 0.081\% = \varepsilon_E$ and $\sigma_S = 82.3\,\text{MPa} < \sigma_E$. This distinction between primary and secondary load responses was previously illustrated in figure 2 and is reproduced here in figure 7. However, beam 3 (point **SP**), *although subjected to a secondary load*, exhibits an intermediate stress state between the purely secondary and purely primary configurations. Point SP lies between points S and P on the elastoplastic stress-strain curve: $\varepsilon_S < (\varepsilon_{PS} = 0.16\%) < \varepsilon_P$ and $\sigma_S < (\sigma_{SP} = 95.85\,\text{MPa}) < \sigma_P$.

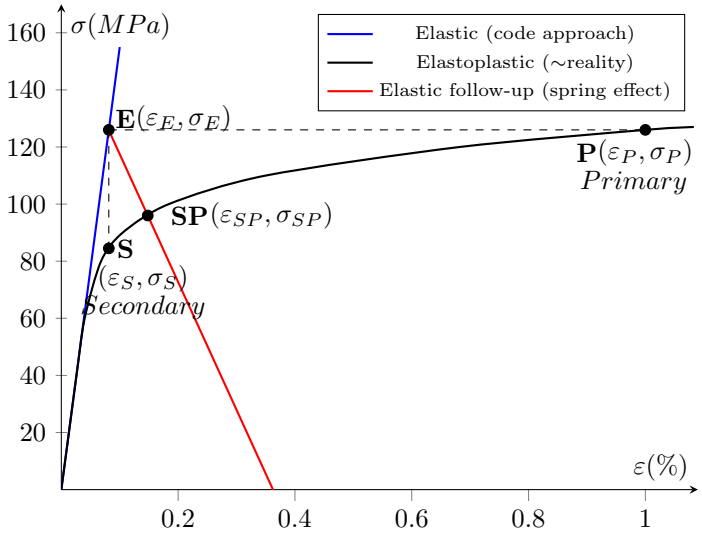


Figure 7: Elastic follow-up – deviation of secondary stress observed in beam 3

Beam 3 in the above example might be understood as a symbolic representation of a piping structure. The common section acts as the source of plasticity due to stress concentrations occurring at specific weak points within the structure. In reality, these vulnerable locations are not only subjected to direct local loading, but are also influenced by loads transfer through the surrounding parts of the piping system. These less vulnerable regions experience lower stress levels and therefore remain entirely within the elastic regime. In this context, the behavior of the weak links responding to the imposed-secondary loads might be deviated compared to the case when no intermediate structure is present, based on how the intermediate structure (spring) elastically responds to the secondary loads.

In the example shown in figure 7, the spring section has a relatively small stiffness compared with the one of the common section, i.e., $k_{spring} = \frac{1}{4} k_{section}$. This stiffness k_{spring} might be viewed as representing a long adjacent piping segment containing several loops, which significantly reduces its overall stiffness. It is also worth noting that the spring section imposes a boundary condition for the common section. Thanks to the surrounding pipings' flexibility, plastic deformation is more likely to develop in the weak zone (the common section). As a result, the position of point **SP** deviates from its initially predicted location (point **S**) under the applied secondary load. In other words, in the presence of spring effect and plasticity, the secondary load is no longer strictly displacement-controlled, as assumed in the methodologies adopted by many conventional piping design codes.
Moreover, if the spring stiffness is almost zero, plastic deformation can develop almost freely in the weak zone. As a result, point **SP** converges toward the position of point **P** (the purely primary-load case), and Beam 3 exhibits essentially the same response as Beam 1. Conversely, if the surrounding structure is very stiff (with an almost infinite stiffness), the weak zone is strongly constrained and plasticity develops in a more controlled manner. In this case, point **SP** approaches the position of point **S** (the purely secondary-load case) on the elastoplastic curve, and Beam 3 exhibits essentially the same response as Beam 2.

The above remarks are illustrated by points B and D in Figure 8. For point B, the spring stiffness is increased to $k/2$, where k is the stiffness of the common section. The imposed displacement is increased to 2.435 mm to achieve the same elastic solution as that predicted for point E. As expected, compared with point SP, the higher spring stiffness causes point B to move closer to the secondary-behavior limit (point S), yielding $\sigma_B = 90.3\,\text{MPa}$ and $\varepsilon_B = 0.13\%$. Similarly, for point D, the initial spring stiffness is reduced to $k/16$, and the imposed displacement is adjusted to 27.591 mm. As expected, point D moves closer to the primary-behavior limit (point P), yielding $\sigma_D = 106.7\,\text{MPa}$ and $\varepsilon_D = 0.283\%$. This parametric study also shows that purely primary behavior is approached as the spring stiffness tends to zero, whereas purely secondary behavior is approached in the limit of infinitely stiff springs.

The three-beam example (Figures 7 and 8) illustrates the limitations of the conventional secondary-stress concept in the presence of elastic follow-up effects. The spring stiffness, represented in Figure 8, significantly influences the elastoplastic response of the weak links. However, spring stiffness itself is not a general parameter for quantifying elastic follow-up in complex piping systems. A more fundamental measure is required to characterize

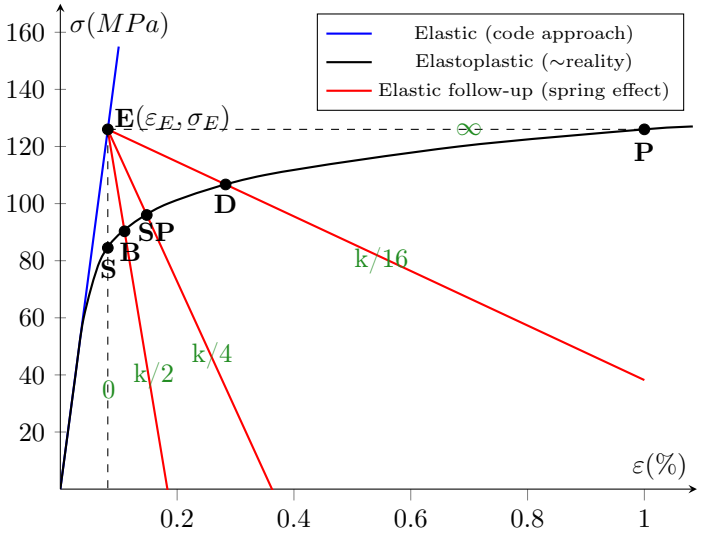


Figure 8: Parametric analysis on the spring stiffness' effect

the phenomenon independently of a particular mechanical idealization. In the next section, the concept of elastic follow-up is revisited from a more general perspective, and a suitable quantification method is introduced.

2.2 Elastic follow-up factor in RCC-MRx

Introduction on RCC-MRx here, without forgetting to mention the nuclear context (which is the focus of this journal publication)
Talk about high temperature pipings and the need to properly consider plasticity

RCC-MRx is a French nuclear design code entitled *Design and Construction Rules for Mechanical Components of Nuclear Installations: High-Temperature, Research and Fusion Reactors*. In the negligible-creep regime, RCC-MRx RB 3650 considers the interaction between elasticity and plasticity under spring effects and quantifies elastic follow-up using the rules provided in RB 3643.3. Displacement-controlled moments are included in the validation of primary stresses in accordance with the requirements of RB 3651 through an abatement coefficient, g, which accounts for the elastic follow-up behavior of the entire structure.

In this paragraph, we focus on the theoretical background of the elastic follow-up factor proposed by Roland L. Roche and subsequently adopted in RCC-MRx RB 3643.3. Finally, the abatement coefficient used in the validation of $S_m + S_b$ stresses is introduced.

This section continues the discussion presented in §2. With the objective of generalizing the elastic follow-up concept and quantifying the degree of elastic follow-up in a piping system subjected to loading, Roland L. Roche introduced two fundamental assumptions.

The above equation of rM is valid for isotropic materials. Considering $\nu = 0.3$ for steels, we have:

$$rM = 0.867\left(\frac{T}{t} - 1\right) \tag{5}$$

In a conservative manner, the code RCC-MRx considers $\frac{2(1+\nu)}{3} \approx 1$, and forces an additional stress indice of 0.87 for pressure and torsional moments, in order to consider the isotropic 3D character of the material. So finally we have the following formula used in RCC-MRx:

$$\mathbf{rM} = \frac{\mathbf{T}}{\mathbf{t}} - \mathbf{1} \tag{6}$$

Note that the higher rM, more the load is considered as having more primary characters. It means that rM(coeff=1) > rM(coeff=0.867) is a conservative approach.

Additionally, based on the definition of local resilience t, we have:

$$T = \frac{\iiint \sigma_0^* \varepsilon_0^* dV}{\iiint \sigma_0^* \varepsilon_0^{p*} dV} = \frac{\iiint \sigma_0^{*2} dV}{\iiint \frac{\sigma_0^{*2}}{t} dV}$$

(assuming that the young modulus is homogeously constant). Finally, if the equivalent stresses are constant in each piping section, we can find the formula of T suggested by the code RCC-MRx:

$$T = \frac{\iiint \sigma_0^{*2} dV}{\iiint \frac{\sigma_0^{*2}}{t} dV} = \frac{\int \sigma_0^{*2} A ds}{\int \frac{\sigma_0^{*2}}{t} A ds}$$

Now, define the elastic follow-up factor as below:

$$rM = \frac{E(\varepsilon^* - \varepsilon_0^*)}{\sigma_0^* - \sigma^*} \tag{7}$$

Up to now, we can find essential formulas used in calculations of rM, from Roland Roche's article, and RCC-MRx.

which defines the global resilience T purely from the elastic solution, without solving the non-linear problem. Passing to equivalent (von Mises) stress and strain increments $\Delta\varepsilon^*$, $\Delta\sigma^*$ as in Eq. (**??**), and using the elastic relation $\Delta\varepsilon_0^* = 2(1+\nu)\,\Delta\sigma_0^*/3E$, Eq. (**??**) reduces to the compact result derived in full in Eq. (**??**) of Appendix B,

$$\frac{3}{2(1+\nu)} \cdot \frac{E\,(\Delta\varepsilon^* - \Delta\varepsilon_0^*)}{\Delta\sigma_0^* - \Delta\sigma^*} = \frac{T}{t} - 1, \tag{8}$$

i.e. the same elastic follow-up factor r_M recovered below from Roche's total-quantity (rather than incremental) notation; the full step-by-step derivation, including the definition of the local resilience t, is reproduced in Appendix B.

RCC-MRx clause RB 3643 formalises the elastic follow-up factor r_M from an energy balance of the purely elastic calculation, following R. Roche's original derivation (reproduced in full in Appendix B). Let $(\sigma_{\mathrm{el}}, \varepsilon_{\mathrm{el}})$ be the elastically computed equivalent stress and strain at the section of interest, and (σ, ε) the actual (elasto-plastic) equivalent stress and strain reached at the same section once the true, non-linear material law (here, Ramberg–Osgood, Eq. (4)) is accounted for, with the loading history assumed radially proportional ($\bar{\bar{\sigma}} = \varphi\,\bar{\bar{\sigma}}_0$, i.e. the elasto-plastic stress tensor field remains everywhere proportional to the elastic one through a single scalar φ). RCC-MRx defines the elastic follow-up factor as

$$r_M \;=\; -\frac{E(\varepsilon - \varepsilon_{\mathrm{el}})}{\sigma - \sigma_{\mathrm{el}}} \;=\; \frac{T}{t} - 1 \;\geq 0, \tag{9}$$

where t is the *local* resilience (degree of elasticity) of the section under consideration, and T is a *global* resilience computed over the whole structure by weighting every section by its elastic reference stress:

$$T = \frac{\int \sigma_{\mathrm{ref}}^2 A\, ds}{\displaystyle\int \frac{\sigma_{\mathrm{ref}}^2 A}{t}\, ds}, \qquad \frac{1}{t} = \frac{\varepsilon_{M,p}(\sigma_{\mathrm{ref}})}{\sigma_{\mathrm{ref}}/E}, \qquad \sigma_{\mathrm{ref}}^2 = \left[0.79\, B_2\, \frac{(m)'_R}{Z}\right]^2 + \left[0.87\, \frac{m_1}{Z}\right]^2, \tag{10}$$

with $\varepsilon_{M,p}(\sigma) = 0.01\left[\sigma/\left(C_0\, R_{p0.2,\mathrm{mean}}\right)\right]^{1/n_0}$ the plastic part of the mean Ramberg–Osgood curve (Eq. (4)), and m_1, $(m)'_R$ the primary and secondary (resultant) moments at the section, Z its section modulus. Equation (10) is the practical, piping-code form of T; its rigorous derivation from the energy balance $\iiint \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}\, dV = \iiint \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}_0\, dV$ is given in Appendix B, Eqs. (13)–(14) and (22).

For isotropic materials, the rigorous derivation of Appendix B gives $r_M = \frac{2(1+\nu)}{3}\left(\frac{T}{t} - 1\right)$, which for steels ($\nu = 0.3$) gives $r_M = 0.867\left(\frac{T}{t} - 1\right)$. Conservatively, RCC-MRx takes $\frac{2(1+\nu)}{3} \approx 1$ (absorbing the 3-D correction into a separate stress index of 0.87 applied to pressure and torsional moments), so that the code ultimately uses the simplified form of Eq. (9), $r_M = T/t - 1$.

2.3 Abatement coefficient g in RCC-MRx

For a section without internal pressure, RCC-MRx clause RB 3651.1131 defines the abatement coefficient as the ratio $g = \sigma/\sigma_{\mathrm{el}}$ of the true elasto-plastic stress to the elastically computed stress at the same imposed strain. The same result can be written directly from r_M and the Ramberg–Osgood law as

$$g \;=\; \frac{\sigma\,(1+r_M)}{E\,\varepsilon_{M,p}(\sigma)+\sigma\,(1+r_M)}. \tag{11}$$

When a primary membrane stress from pressure, $\sigma_P = 0.87\,\dfrac{PD_i}{2h_c}$, is also present at the section (with $\sigma_P < \sigma$), the code generalises Eq. (11) to

$$g \;=\; \frac{(\sigma-\sigma_P)(1+r_M)}{E\,[\varepsilon_{M,p}(\sigma)-\varepsilon_{M,p}(\sigma_P)]+(\sigma-\sigma_P)(1+r_M)}. \tag{12}$$

Figure 9 reproduces the code's own graphical construction of g, equivalent to Eqs. (11)–(12).

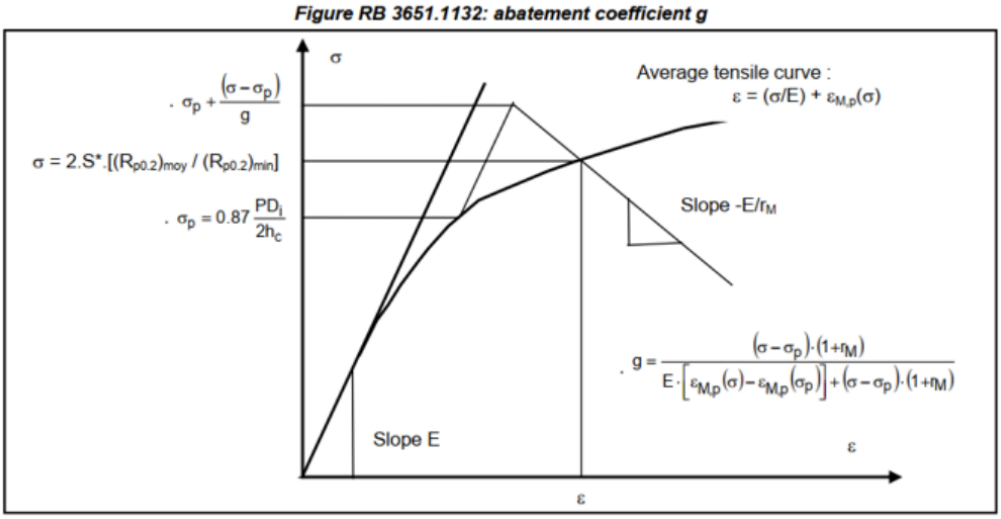


Figure 9: RCC-MRx Figure RB 3651.1132 (reproduced from the code): graphical definition of the abatement coefficient g, showing the average Ramberg–Osgood tensile curve, the elastic (slope E) and elastic-follow-up (slope $-E/r_M$) construction lines, the primary membrane stress σ_p and allowable stress S^*.

The coefficient g is then used to reduce ("abate") the displacement-dependent moment m (secondary, in the classical sense) into an abated value $g\,m$, which is combined with the primary loads in the assessment of primary-type (P-type) damage through the RCC-MRx CMS1 stress-limitation rule (RB 3651.1131):

$$\overline{s_m + s_b} = \left[\left(D_1\left[\frac{PD_i}{2h_c}\right]\right)^2 + \langle D_2/Z,\; M + g\cdot m\rangle^2\right]^{0.5} \le S^*. \tag{13}$$

In this way, the combined effects of elastic follow-up and plasticity are accounted for in the P-type damage check of RCC-MRx, via $\overline{s_m + s_b}$, without requiring the analyst to perform a full elasto-plastic finite-element calculation. The presentation motivates exactly this rule by asking whether secondary loads must be considered when validating primary stresses: the answer is illustrated by re-using the Bar 1/Bar 2/Bar 3 Ramberg–Osgood curve already given in Figure 21 (bottom-right panel) – since Bar 3's response (points A–D) sits strictly between the purely primary (Bar 1) and purely secondary (Bar 2) limits, a nominally secondary moment can behave partly as a primary one, which is precisely why $g \cdot m$, rather than m alone, is added inside the CMS1 check of Eq. (13).

For the benchmark section reported later in Figure **??**, the code output gives $\sigma = 119.59\,\mathrm{MPa}$, $\varepsilon_{M,p}(\sigma) = 0.587\,\%$ and $r_M = 2.638$. Substituting these values into Eq. (11) with $E = 155\,000\,\mathrm{MPa}$ yields

$$g = \frac{119.59(1+2.638)}{155000\times 0.00587 + 119.59(1+2.638)} = 0.3235,$$

i.e. the reported production-code value $g = 0.3237$ to within rounding of the tabulated strain $\varepsilon_{M,p}$. This direct numerical check was used here as an internal verification of the transcribed RCC-MRx formulae.

Table 2: Consolidated notation for the RCC-MRx elastic-follow-up route.

Quantity	**Definition / role**
t	Local resilience of the section, defined through $1/t = \varepsilon_{M,p}(\sigma_{\text{ref}})/(\sigma_{\text{ref}}/E)$; low t means a locally more plasticised section.
T	Global resilience of the whole structure, obtained by stress-squared weighting of all sections through Eq. (10).
r_M	Elastic follow-up factor, $r_M = T/t - 1$ in the practical RCC-MRx form (or $r_M = \frac{2(1+\nu)}{3}(T/t-1)$ in the rigorous isotropic derivation reproduced in Appendix B).
g	Abatement coefficient applied to the displacement-controlled moment, given by Eq. (11) without pressure and by Eq. (12) when a primary membrane pressure stress σ_P is present.
σ_{ref}	Elastic reference equivalent stress used to build T and t; for piping sections it is assembled from the weighted primary and secondary moment terms in Eq. (10).

Table 2 is intended as a quick-reference summary; its notation is used consistently throughout the remainder of the paper and in Appendix B.

This raises a natural verification question, addressed by the benchmark of this paper: *is it necessary to explicitly reconsider secondary loads in the primary-stress validation, and if so, does the RCC-MRx analytical route (r_M, g) correctly and conservatively capture the elasto-plastic reality?*

Talk about sm+sb + type P damages here – including elastic and elastoplastic instability

Maybe we can say that the above method uses a global approach for the calculation of rM –> g accounted in sm+sb, to cover a global damage mode which is elastic/elastoplastic instability

We'll need the RCC-MRx criteria to understand more on the allowble stresses of sm+sb in the code

We might talk more about the fact that elastic follow-up might also be used in local analysis – see ASME III NB for fatigue and progressive deformation - correction factor

Maybe we can discuss on the topic of g unique here, based on Katchanov assumption of Roland Roche

3 Elasto-plastic benchmark on the abattement coefficient g

Some introduction here
says we review the Roland Roche approach by a FEA elastoplastic large strain model
Explain that the same model as the elastic benchmark is used here, with non-linear material lawl

3.1 Elastoplastic large-strain FEA analysis vs PEPS model

Use the same model as the elastic benchmark + plasticity

The elasto-plastic shell analysis uses the von Mises yield criterion with isotropic hardening, and the Ramberg–Osgood law of Eq. (4) calibrated to 316L(N) at 550 °C (RCC-MRx RB A3.3S.451), with yielding onset defined at zero offset stress consistent with the code's convention. The material is defined in MSC Marc by $E = 155\,000$ MPa, $\nu = 0.3$, and the von Mises yield condition $\sigma_{\mathrm{VM}} < \sigma_Y$ with

$$\sigma_{\mathrm{VM}} := \sqrt{\tfrac{3}{2}\, s_{ij} s_{ij}}, \tag{14}$$

s_{ij} the deviatoric stress. Hardening is tracked through the equivalent plastic strain

$$\varepsilon_{p,\mathrm{eq}} := \sqrt{\tfrac{2}{3}\, \varepsilon^p_{ij} \varepsilon^p_{ij}}\,, \tag{15}$$

with ε^p_{ij} the plastic strain tensor, consistent with the standard von Mises formulation. Isotropic hardening is imposed as a tabulated function $\sigma_Y(\varepsilon_{p,\mathrm{eq}}) := F(V1)$, with $V1$ the independent (equivalent plastic strain) variable digitised from the Ramberg–Osgood law of Eq. (4) at zero offset stress: the presentation's *plas_new* table samples this curve over 34 points, $V1 \in [0, 1.002]$, $F \in [0.6, 213.2]$ MPa (dimensionless scale factor $\times 100$), reproduced in Figure 10.

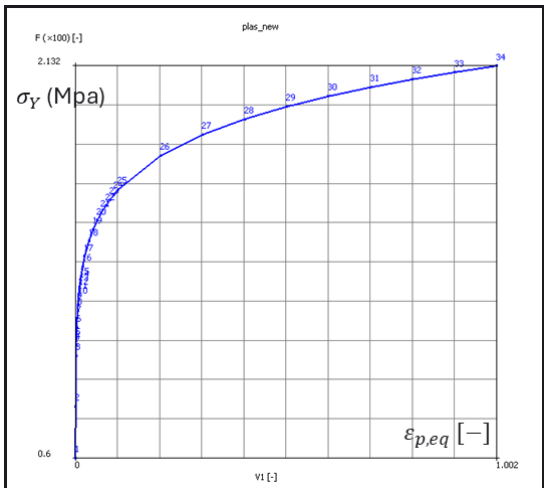


Figure 10: MSC Marc tabulated isotropic hardening curve (*plas_new* table), σ_Y versus equivalent plastic strain $\varepsilon_{p,\mathrm{eq}}$, digitised at 34 points from the zero-offset Ramberg–Osgood law of Eq. (4) and used directly as the FE material input. –> higher quality figure to replace

The shell mesh used for the elasto-plastic run (80 elements around the circumference) was checked beforehand for consistency with the elastic reference solution of Section **??**. The source presentation makes this consistency check explicit by reading the retained critical-point values back onto the FE hardening law: the final shell result combines $\sigma_{\mathrm{VM}} = 86.5$ MPa with an equivalent plastic strain of about $\varepsilon_{p,\mathrm{eq}} = 0.05\,\%$ (Eq. (15)), and the corresponding point lies on the imposed work-hardening curve $\sigma_Y(\varepsilon_{p,\mathrm{eq}})$ used in the Marc material definition. Figure 11 zooms into the small-strain region of this same curve (equivalent plastic strain up to $0.12\,\%$) and marks this exact point, directly confirming graphically that the FE critical-section result is consistent with the imposed hardening law.

The von Mises stress distribution over the elbow shows a clearly identifiable "weak link" region (the section of maximum bending combined with ovalisation-amplified local stress), consistently located at the same position across all models, which is used as the reference cross-section for the abatement-coefficient comparison below. Figure 12 gives the full-field response, while Figure 13 zooms in on the section from which the FE-based value of g is back-calculated.

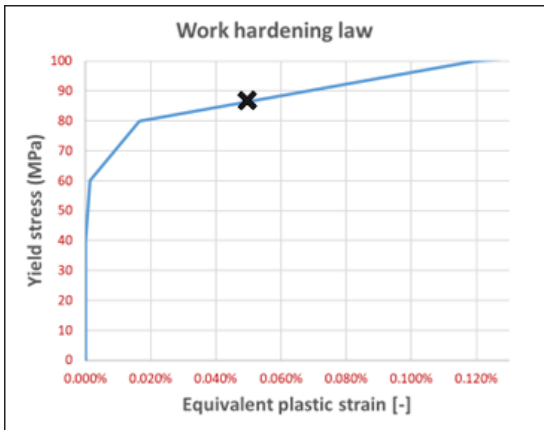


Figure 11: Zoom of the work-hardening law over the small-strain range relevant to the benchmark, with the retained critical-section point ($\sigma_{\mathrm{VM}} = 86.5\,\mathrm{MPa}$, $\varepsilon_{p,\mathrm{eq}} = 0.05\,\%$) marked, confirming its consistency with the imposed FE hardening curve. –> figure to remove

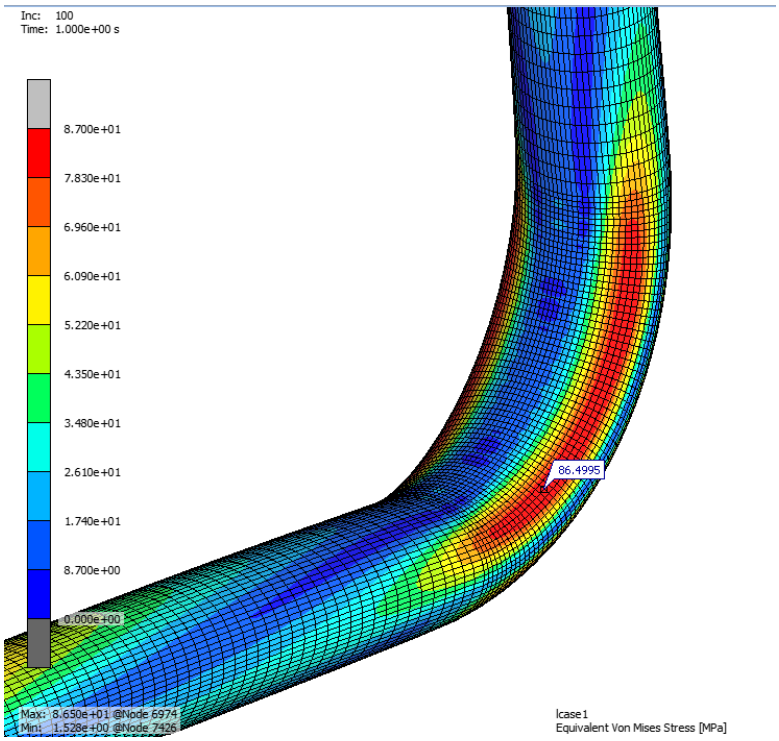


Figure 12: MSC Marc elasto-plastic shell model (80 elements/circumference): equivalent von Mises stress distribution at the end of the thermal transient, showing the critical section at the elbow crown (maximum 86.5 MPa, node 6974; minimum 1.5 MPa, node 7426).

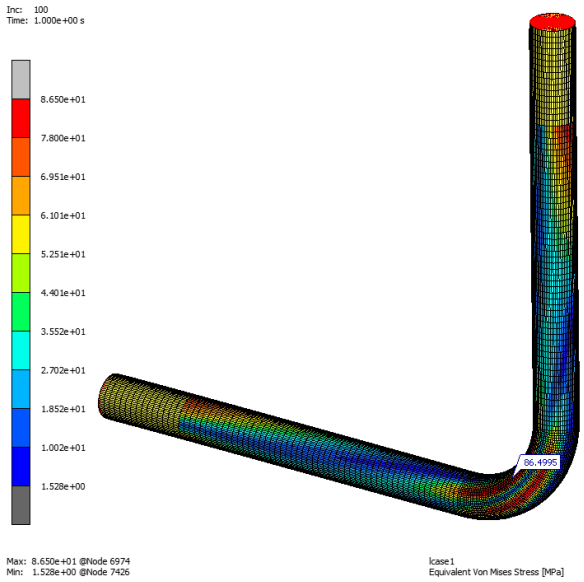


Figure 13: Close-up of the critical elbow section used for the FE/RCC-MRx comparison: the local maximum equivalent von Mises stress of 86.5 MPa is the reference value from which the FE-based abatement coefficient is back-calculated.

3.2 Observations and discussions

The RCC-MRx elastic route (RB 3651.1131) reduces the secondary (displacement-dependent) moment m by the coefficient g and combines the abated stress with the primary stress in the $\sigma_m + \sigma_b$ (P-type) damage check; equivalently, g can be interpreted either as an abatement applied to the stresses (from the elastic/purely-secondary value down to the effective value) or as an abatement applied to the moments (from the elastic moment down to the effective moment). The natural benchmark question is then: *how does the RCC-MRx-derived g (and the associated $\sigma_m + \sigma_b$) compare with the coefficient and stress obtained from a full elasto-plastic FE reference calculation at the same critical section?*

Table 3 reports this comparison.

Table 3: Abatement coefficient g and equivalent stress: RCC-MRx analytical route versus elasto-plastic FE reference.

Model / route	g	**Stress [MPa]**
Aspect Nuclear Pipe Stress, RCC-MRx (SMC1), $\sigma_m + \sigma_b$	0.323	66.5
MSC Marc elasto-plastic shell, reaction-based effective abatement	0.436	86.5

Taken together, Figures **??** and 14 provide the production-code evidence behind the benchmark values reported in Table 3: the first figure documents the local r_M/g calculation, while the second gives the corresponding stress result at the critical point. The source presentation additionally distinguishes two FE-side ways of defining an "effective" abatement coefficient from the shell model: a local stress ratio, $g_{\mathrm{VM}} = 86.5/213.7 = 0.405$, and a reaction-moment ratio, $g_{\mathrm{reaction}} = 17.58/40.3 = 0.436$, reconstructed independently in Figure 15. The presentation retains the second value in the comparison table because RCC-MRx applies g to the displacement-dependent moment rather than directly to the local von Mises stress.

Two observations follow directly from Table 3:

1. **Observation 1:** the RCC-MRx abatement coefficient ($g = 0.323$) is less conservative than the coefficient implied by the elasto-plastic FE reference ($g = 0.436$).
2. **Observation 2:** the RCC-MRx primary-equivalent stress ($\sigma_m + \sigma_b = 66.5$ MPa) is less conservative than the von Mises stress of the reference elasto-plastic FE analysis (86.5 MPa).

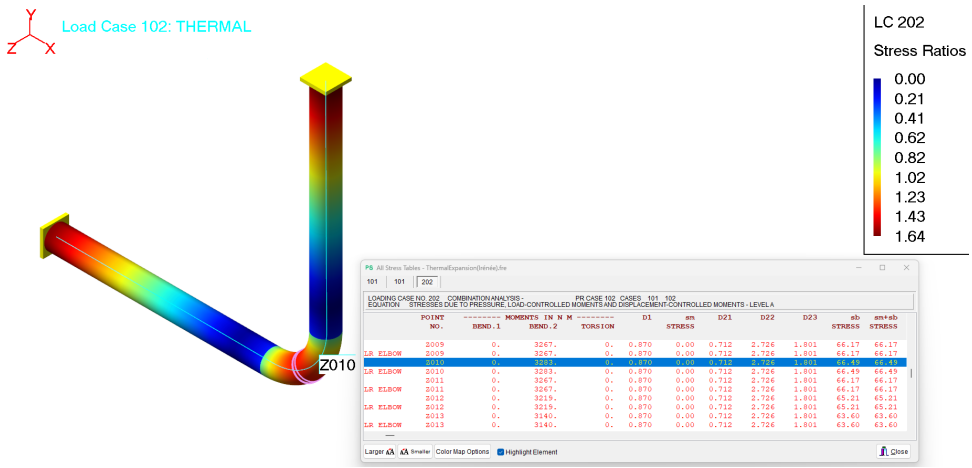


Figure 14: Aspect Nuclear Pipe Stress (NS solver) results for the thermal expansion load case (LC 202): stress-ratio colour map along the model and detailed stress table at the elbow points (Z009–Z013), giving $\sigma_m + \sigma_b = 66.49\,\text{MPa}$ at the critical point Z010, consistent with Table 3.

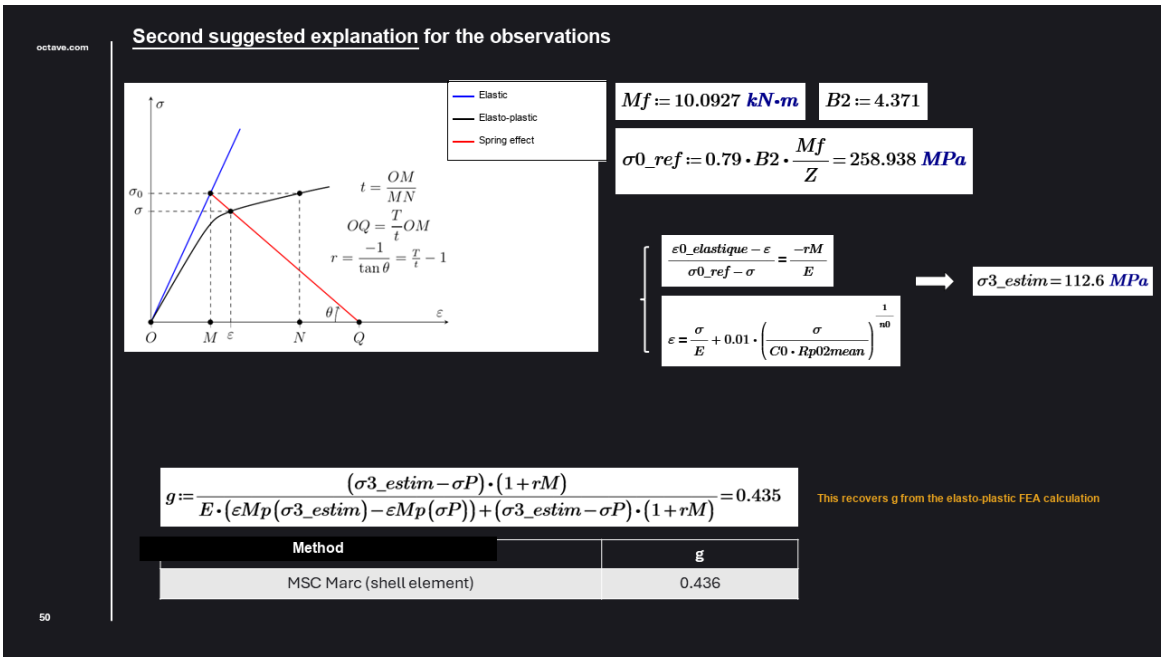


Figure 15: Second FE-side consistency check reproduced from the presentation: starting from the shell-model reaction moment $M_f = 10.0927\,\text{kN m}$ and the piping-code coefficient $B_2 = 4.371$, the equivalent elastic reference stress is reconstructed as $\sigma_{0,\text{ref}} = 0.79\,B_2\,M_f/Z = 258.938\,\text{MPa}$. Solving the coupled elastic-follow-up and Ramberg–Osgood relations shown on the slide then gives $\sigma_{3,\text{est}} = 112.6\,\text{MPa}$ and $g = 0.435$, i.e. the reaction-based FE abatement coefficient $g_{\text{reaction}} = 0.436$ to within rounding. –> to redesign

Discussion: why is the RCC-MRx route less conservative?

Slide 44 of the source presentation first recalls clause RB 3651.111, titled (in the original French code) *Limitations des contraintes dues à la pression* ("Limitation of stresses due to pressure"), which gathers the classical primary-pressure limits

$$P_m \le S_m(\theta_m), \qquad P_L \le 1.5\, S_m(\theta_m), \qquad P_L + P_b \le 1.5\, S_m(\theta_m).$$

Within that same elastic-analysis framework, the subsequent conversion argument then moves to RB 3651.112, i.e. the general primary membrane-plus-bending check used here. RCC-MRx clause RB 3651.112 requires that, for load combinations governed by level-A criteria, the general primary membrane-plus-bending stress additionally satisfies

$$\overline{p_m + p_b} = B_1\, P \left[\frac{D_e}{2h_c}\right] + \langle B_2/Z,\, M\rangle \le 1.3\, S_m(\theta_m), \tag{16}$$

i.e. the abated, displacement-dependent stress must ultimately be re-expressed and re-checked on the strictly primary allowable basis $1.3\, S_m(\theta_m)$, rather than on the CMS1 allowable S^* of Eq. (13) used to derive g in Section 2.2. For the benchmark section considered here, $1.3\, S_m = 100.1\,\text{MPa}$ while $S^* = 48.156\,\text{MPa}$, so that the ratio

$$\frac{1.3\, S_m}{S^*} = \frac{100.1}{48.156} = 2.079 \tag{17}$$

is the conversion factor that must be applied when re-scaling the RCC-MRx abatement coefficient and stress from the S^* (CMS1) basis onto the $1.3\, S_m$ (primary) basis:

$$\begin{aligned} g^{\text{rabat}}_{\text{RCC-MRx}} &= g \cdot \frac{1.3\, S_m}{S^*} = 0.323 \times 2.079 = 0.671, \\ (\sigma_m + \sigma_b)^{\text{rabat}} &= (\sigma_m + \sigma_b) \cdot \frac{1.3\, S_m}{S^*} = 138.23\,\text{MPa}. \end{aligned} \tag{18}$$

Let's review the above paragraph to see if it's appropriate to keep it in the article

Once this conversion is applied, the RCC-MRx-based primary-equivalent stress ($g^{\text{rabat}} = 0.671$, $(\sigma_m + \sigma_b)^{\text{rabat}} = 138.23\,\text{MPa}$) becomes markedly *more* conservative than the elasto-plastic FE reference ($g = 0.436$, $\sigma_{\text{VM}} = 86.5\,\text{MPa}$), reversing the apparent non-conservatism identified in Table 3. This confirms that the raw comparison of Table 3 is not made on a consistent allowable-stress basis: g and $\sigma_m + \sigma_b$ as computed under the CMS1 rule (RB 3651.1131, allowable S^*) are not directly comparable to a von Mises stress checked against $1.3\, S_m$, and the code-mandated re-scaling of Eq. (16) restores the expected conservative character of the analytical RCC-MRx route.

A complementary explanation emerges from the mesh sensitivity already noted in Table 1. Two distinct pieces of evidence are available in the source presentation. First, slide 38 contains a native Office chart titled *Mesh convergence analysis*, whose exact data are reproduced in Appendix **??** (Table **??**). Those four points are the maximum von Mises stresses extracted from the shell calculations, not the g values themselves: the raw stress sequence is 105.02 MPa, 83.86 MPa, 84.57 MPa and 86.5 MPa for 10, 20, 40 and 80 circumferential elements, respectively, so it is not strictly monotonic. The key observation is that the coarse 10-element mesh is an outlier, whereas the 20-, 40- and 80-element meshes cluster within a much narrower band and approach the retained elasto-plastic reference value of 86.5 MPa. Second, the separate NS/NewSolver sensitivity plot reproduced in Figure **??** tracks the derived RCC-MRx abatement coefficient as the discretisation is refined; it shows an overall upward stabilisation trend with small oscillations, rather than a perfectly monotonic increase. Taken together, these two slides indicate that once mesh convergence is properly achieved, the discrepancy between the RCC-MRx analytical route and the elasto-plastic reference is substantially reduced, even before the primary-equivalent stress conversion of Section **??** is applied. This suggests that part of the discrepancy originally observed in Table 3 originates from an insufficiently refined shell mesh in the reference elasto-plastic model, rather than from an intrinsic shortcoming of the RCC-MRx r_M/g formulation itself.

Both explanations are consistent with one another and are not mutually exclusive: the primary-equivalent stress conversion is a code-mandated, conservatism-restoring step that must always be applied; the mesh-sensitivity finding is a numerical-modelling caveat that must be controlled independently, regardless of which analytical route (RCC-MRx or direct elasto-plastic FE) is used.

4 Conclusions

1. The paper provides a consolidated route from the historical background and constitutive assumptions to the practical RCC-MRx definitions of r_M and g, supported by the original diagrams, code excerpts and derivation material from the source presentation.
2. The elastic benchmark shows that the beam-element piping model and the MSC Marc shell model agree well in terms of global flexibility once the elbow flexibility factor is introduced consistently and the boundary conditions are treated carefully.
3. For the benchmark elbow, the direct RCC-MRx output gives $r_M = 2.638$ and $g = 0.3237$, whereas the non-linear shell reference yields a higher effective abatement level when the comparison is made on a raw section-stress basis.
4. This apparent discrepancy is resolved by two complementary observations: the code requires the abated secondary effect to be expressed in terms of primary stress, and the shell reference solution is also sensitive to mesh density in the elbow ovalisation zone (see also Appendix **??**).
5. The additional wall-thickness variants and the cantilever validation case support the robustness of the implementation and show that the analytical elastic follow-up route can be checked independently of the main elbow benchmark.
6. Modern numerical tools make such comparisons much faster to perform, but they also increase the importance of modelling choices and verification steps, especially those related to mesh density, shell formulation and boundary-condition idealisation.
7. The present work remains limited by the adopted boundary conditions, shell formulation, temperature-invariant Ramberg–Osgood shape parameters, and the omission of creep, stress relaxation and pressure effects. Extending the study to include pressure loading, alternative hardening descriptions and further mesh-convergence work is a natural next step.
8. Several open questions regarding the domain of validity of this simplified, temperature-invariant hardening shape are noted in the presentation (and are only partially addressed here): the sensitivity to strain rate, to monotonic versus cyclic hardening, to the number of cycles, and to environment (air, PWR primary water, sodium, etc.). Figure 16 documents both the temperature family and the parameter sensitivity used to motivate this discussion. For reference, an example comparison at 0.01 true strain shows a stress discrepancy of about 4.0 MPa (∼5 %) between candidate hardening curves [7].

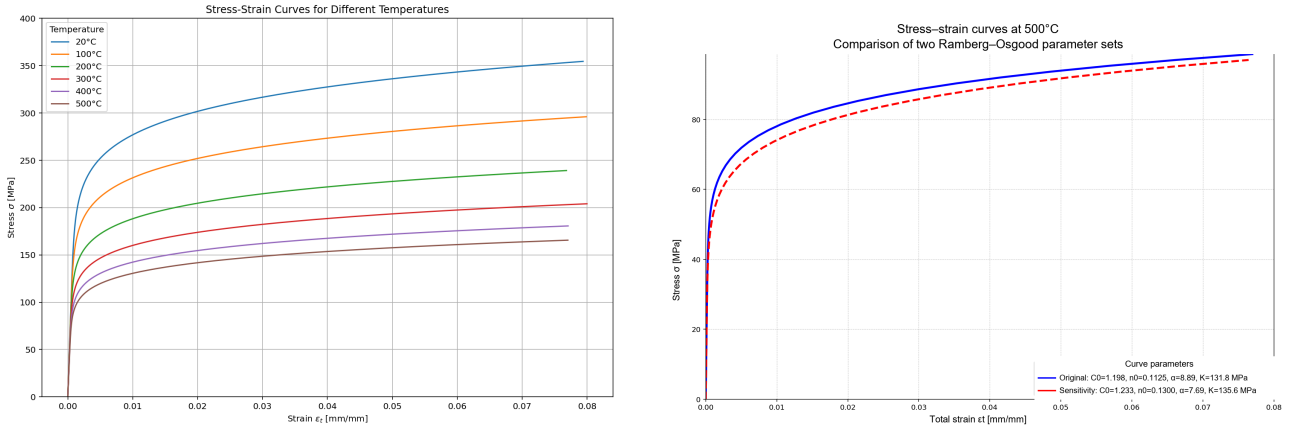


Figure 16: Ramberg–Osgood material data extracted from the presentation. Left: family of 316L(N) stress–strain curves used to justify the temperature-dependent scaling of the law through $R_{p0.2}(T)$. Right: sensitivity check at 500 °C (a different temperature from the 550 °C benchmark), comparing the reference parameter set ($C_0 = 1.198$, $n_0 = 0.1125$, $C_0 R_{p0.2} \approx 131.8$ MPa) with the alternative fit ($C_0 = 1.233$, $n_0 = 0.130$, $C_0 R_{p0.2} \approx 135.6$ MPa) discussed in the text. –>show the importance of n0/C0 on the final results

Acknowledgements

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Octave teams + Cadence

A Definition of elastic follow-up factor

The first assumption pertains to loading-path nonlinearity and is based on Kachanov's hypothesis of a radially proportional (self-similar) stress redistribution:

$$\bar{\bar{\sigma}} = \varphi\, \bar{\bar{\sigma}}_0 \tag{19}$$

Equation (19) might also be expressed for the equivalent stresses:

$$\sigma^* = \varphi \sigma_0^* \tag{20}$$

$(\Delta\bar{\bar{\varepsilon}}_0, \Delta\bar{\bar{\sigma}}_0)$ denotes the strain–stress tensor state of the material point obtained from a purely elastic analysis. ε_0^* and σ_0^* are the corresponding equivalent elastic strain and stress, respectively (schematically represented by point **E** in Figure 7). See appendix D for further details.

$(\Delta\bar{\bar{\varepsilon}}, \Delta\bar{\bar{\sigma}})$ denotes the strain–stress tensor state of the material point obtained from an elastoplastic analysis. ε^* and σ^* are the corresponding equivalent elastoplastic strain and stress, respectively (schematically represented by point **SP** in Figure 7).

The second assumption concerns material nonlinearity, represented by the Ramberg–Osgood law. RCC-MRx provides the corresponding uniaxial stress–strain relation as follows:

$$\varepsilon_{axial} = \frac{\sigma_{axial}}{E} + \left(\frac{\sigma_{axial}}{K}\right)^{1/n} \tag{21}$$

A.1 First approach: Ramberg-Osgood relation under uniaxial configurations

In this first approach, we consider only the axial deformation of the beam under an imposed displacement **u**. All stress and strain tensors are represented by their corresponding axial scalar quantities: **SP**(ε, σ) and **E**(ε_0, σ_0).

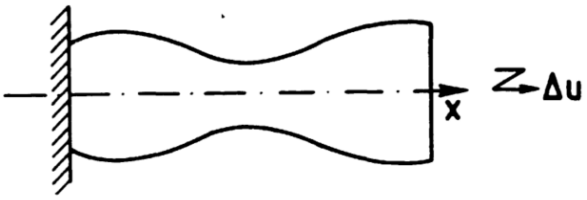


Figure 17: A beam is axially loaded by a secodary load-to adap the figure: cycle displacement deltaU is out scope

For uniaxial loads, the RCC-MRx Ramberg-Osgood formula can be directly applied:

$$\varepsilon(x) = \frac{\sigma(x)}{E} + \left(\frac{\sigma(x)}{K}\right)^{1/n} \tag{22}$$

The Khachanov assumption is simply $\sigma(x) = \varphi\sigma_0(x)$. The Ramberg–Osgood relation is rewritten as follows:

$$\varepsilon(x) = \frac{\varphi\sigma_0(x)}{E} + \left(\frac{\varphi\sigma_0(x)}{K}\right)^{1/n} = \varphi\varepsilon_0(x) + \phi\varepsilon_0^p(x) \tag{23}$$

where ε_0^p is the equivalent plastic strain obtained from the Ramberg–Osgood law using the elastic stress σ_0, and $\phi = \varphi^n$.

Now, let's analyze the structure using an elastoplastic approach (point **SP**):

$$\int_{x=0}^{x=L} \delta\sigma(x)\varepsilon(x)A(x)\,dx = \delta\sigma(L)A(L)u$$

Using relation (20) and introducing the variation of the unknown field, $\delta\varphi$, yields

$$\int_{x=0}^{x=L} (\delta\varphi)\sigma_0(x)\varepsilon(x)A(x)\,dx = (\delta\varphi)\sigma_0(L)A(L)u$$

$$\int_{x=0}^{x=L} \sigma_0(x)\varepsilon(x)A(x)\,dx = \sigma_0(L)A(L)u \tag{24}$$

For the elastic solution (point **E**), consider the same secondary loads **u**. The principle of virtual work gives

$$\frac{1}{2}\int_{x=0}^{x=L} \sigma_0(x)\varepsilon_0(x)A(x)\,dx = \frac{1}{2}\sigma_0(L)A(L)u \tag{25}$$

The energy balances for the elastic solution (Equation (25)) and the elastoplastic solution (Equation (24)) yield

$$\int_{x=0}^{x=L} \sigma_0(x)\varepsilon(x)A(x)\,dx = \int_{x=0}^{x=L} \sigma_0(x)\varepsilon_0(x)A(x)\,dx$$

Substituting Equations (19) and (23) into the above expression yields

$$\int_{x=0}^{x=L} \sigma_0(x)\left(\varphi(x)\varepsilon_0(x) + \phi\varepsilon_0^p(x)\right)A(x)\,dx = \int_{x=0}^{x=L} \sigma_0(x)\varepsilon_0(x)A(x)\,dx$$

$$(\varphi-1)\int_{x=0}^{x=L} \sigma_0(x)\varepsilon_0(x)A(x)\,dx + \phi\int_{x=0}^{x=L} \sigma_0(x)\varepsilon_0^p(x)A(x)\,dx = 0 \tag{26}$$

Defining the local resilience t and global resilience T as below:

$$\frac{1}{t} = \frac{\varepsilon_0^p(x)}{\varepsilon_0(x)}$$

$$\frac{1}{T} = \frac{\int_{x=0}^{x=L} \sigma_0(x)\varepsilon_0^p(x)A(x)\,dx}{\int_{x=0}^{x=L} \sigma_0(x)\varepsilon_0(x)A(x)\,dx} = \frac{\int_{x=0}^{x=L} \frac{1}{t}\sigma_0^2(x)A(x)\,dx}{\int_{x=0}^{x=L} \sigma_0^2(x)A(x)\,dx}$$

The above equation (26) can be rewritten as

$$(\varphi-1)\,T + \phi = 0 \tag{27}$$

Now, starting from equation (23), we have:

$$\varepsilon(x) = \varphi\varepsilon_0(x) + \phi\varepsilon_0^p(x)$$

$$\varepsilon(x) - \varepsilon_0(x) = \varepsilon_0(x)(\varphi-1)\left(1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^p(x)}{\varepsilon_0(x)}\right)$$

$$\frac{\varepsilon(x)-\varepsilon_0(x)}{(\varphi-1)\,\varepsilon_0(x)} = 1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^p(x)}{\varepsilon_0(x)}$$

$$\frac{E(\varepsilon(x)-\varepsilon_0(x))}{(\varphi-1)\,\sigma_0(x)} = 1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^p(x)}{\varepsilon_0(x)}$$

$$\frac{E(\varepsilon(x)-\varepsilon_0(x))}{\sigma(x)-\sigma_0(x)} = 1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^p(x)}{\varepsilon_0(x)}$$

$$\frac{E(\varepsilon(x)-\varepsilon_0(x))}{\sigma_0(x)-\sigma(x)} = -1 + \frac{T}{t} \qquad \text{(from equation (27))} \tag{28}$$

Now we define the elastic-follow-up factor $rM = \frac{E(\varepsilon(x)-\varepsilon_0(x)}{\sigma_0(x)-\sigma(x)}$, then:

$$rM = \frac{T}{t} - 1 \tag{29}$$

A.2 Second approach: Generalized stress-strain tensors Ramberg-Osgood under multiaxial configurations

In context of a multiaxial configuration, the Ramberg Osgood law is generalized for the plastic deformation tensor:

$$\bar{\bar{\varepsilon}}^p(\bar{\bar{\sigma}}) = \phi(\varphi)\bar{\bar{\varepsilon}}^p(\bar{\bar{\sigma}}_0)$$

$$\bar{\bar{\varepsilon}}^p = \phi(\varphi)\bar{\bar{\varepsilon}}^p_0 \tag{30}$$

The total elastoplastic strain tensor is expressed as below:

$$\bar{\bar{\varepsilon}} = \bar{\bar{\varepsilon}}^{el} + \bar{\bar{\varepsilon}}^p$$

Following Hooke' law $\bar{\bar{\sigma}} = \bar{\bar{C}}\bar{\bar{\varepsilon}}^{el}$ and equation (30), we have:

$$\bar{\bar{\varepsilon}} = \bar{\bar{C}}^{-1}\bar{\bar{\sigma}} + \phi\bar{\bar{\varepsilon}}^p_0 = \varphi\bar{\bar{C}}^{-1}\bar{\bar{\sigma}}_0 + \phi\bar{\bar{\varepsilon}}^p_0 = \varphi\bar{\bar{\varepsilon}}_0 + \phi\bar{\bar{\varepsilon}}^p_0 \tag{31}$$

Equation (31) might also be expressed for the equivalent strains:

$$\varepsilon^* = \varphi\varepsilon^*_0 + \phi\varepsilon^{p*}_0 \tag{32}$$

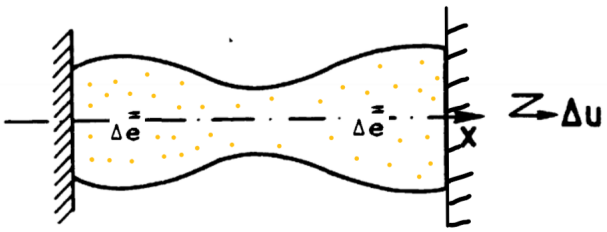


Figure 18: Energy-balance scheme extracted from the presentation, summarising Roche's route from the virtual-work statement of the elastic calculation to the equivalent stress/strain formulation used to define T, t and ultimately r_M.

For the elastoplastic analysis (point **SP**), the local equilibrium equation is given by $\sigma_{ij,j} + f_i = 0$ (with $i, j = \overline{1,3}$). Considering an imposed displacement field $\vec{u}$ prescribed on the boundary of the domain, together with a thermal expansion load $\bar{\bar{\epsilon}}$ acting on all material points (see Figure 18), application of the divergence theorem and the principle of virtual work leads to the following relation involving an infinitesimal variation of the stress field, $\delta\sigma$:

$$\iiint \delta\bar{\bar{\sigma}} : \bar{\bar{\varepsilon}}\, dV = \iint \delta\bar{\bar{\sigma}}\, \vec{n}\, \vec{u}\, dS - \iiint \delta\bar{\bar{\sigma}} : \bar{\bar{\epsilon}}\, dV \tag{33}$$

Using relation (20) and introducing the variation of the unknown field, $\delta\varphi$, yields

$$\iiint_V (\delta\varphi)\, \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}\, dV = \iint_S (\delta\varphi)\, \bar{\bar{\sigma}}_0 \cdot \vec{n} \cdot \vec{u}\, dS - \iiint_V (\delta\varphi)\, \bar{\bar{\sigma}}_0 : \bar{\bar{\epsilon}}\, dV$$

$$\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}\, dV = \iint_S \bar{\bar{\sigma}}_0 \cdot \vec{n} \cdot \vec{u}\, dS - \iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\epsilon}}\, dV \tag{34}$$

For the elastic solution (point **E**), consider the same secondary loads ($\vec{u}$ and $\bar{\bar{\epsilon}}$). The principle of virtual work gives

$$\frac{1}{2}\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}_0\, dV = \frac{1}{2}\iint_S \bar{\bar{\sigma}}_0 \cdot \vec{n} \cdot \vec{u}\, dS - \frac{1}{2}\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\epsilon}}\, dV \tag{35}$$

The energy balances for the elastic solution (Equation (35)) and the elastoplastic solution (Equation (34)) yield

$$\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}\, dV = \iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}_0\, dV,$$
$$\iiint_V \frac{1}{\varphi}\, \bar{\bar{\sigma}} : \bar{\bar{\varepsilon}}\, dV = \iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}_0\, dV$$

Neglecting the contribution of volumetric strains to the strain energy, the above tensorial relation can be expressed in terms of the equivalent stress and equivalent strain as

$$\iiint_V \frac{1}{\varphi}\,\sigma^*\varepsilon^*\,dV = \iiint_V \sigma_0^*\varepsilon_0^*\,dV$$

Substituting Equations (32) and (20) into the above expression yields

$$\iiint_V \sigma_0^*\left(\varphi\varepsilon_0^* + \phi\varepsilon_0^{p*}\right)\,dV = \iiint_V \sigma_0^*\varepsilon_0^*\,dV$$

$$(\varphi-1)\iiint_V \sigma_0^*\varepsilon_0^*\,dV + \phi\iiint_V \sigma_0^*\varepsilon_0^{p*}\,dV = 0 \tag{36}$$

Defining the local resilience t and global resilience T as below:

$$\frac{1}{t} = \frac{\varepsilon_0^{p*}}{\varepsilon_0^*}$$

$$\frac{1}{T} = \frac{\iiint_V \sigma_0^*\varepsilon_0^{p*}\,dV}{\iiint_V \sigma_0^*\varepsilon_0^*\,dV} = \frac{\iiint_V \frac{1}{t}\sigma_0^{*2}\,dV}{\iiint_V \sigma_0^{*2}\,dV}$$

The above equation (36) can be rewritten as

$$(\varphi-1)\,T + \phi = 0 \tag{37}$$

Now, starting from equation (32), we have:

$$\varepsilon^* = \varphi\varepsilon_0^* + \phi\varepsilon_0^{p*}$$

$$\varepsilon^* - \varepsilon_0^* = \varepsilon_0^*(\varphi-1)\left(1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^{p*}}{\varepsilon_0^*}\right)$$

$$\frac{\varepsilon^* - \varepsilon_0^*}{(\varphi-1)\,\varepsilon_0^*} = 1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^{p*}}{\varepsilon_0^*}$$

$$\frac{\varepsilon^* - \varepsilon_0^*}{(\varphi-1)\left(\frac{2}{3}\frac{1+\nu}{E}\sigma_0^*\right)} = 1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^{p*}}{\varepsilon_0^*} \qquad \text{(from equation (49), see Appendix D)}$$

$$\frac{E(\varepsilon^* - \varepsilon_0^*)}{(\varphi-1)\sigma_0^*}\frac{3}{2(1+\nu)} = 1 + \frac{\phi}{\varphi-1}\frac{\varepsilon_0^{p*}}{\varepsilon_0^*}$$

$$\frac{E(\varepsilon^* - \varepsilon_0^*)}{\sigma_0^* - \sigma^*} = 0.867\left(-1 - \frac{\phi}{\varphi-1}\frac{\varepsilon_0^{p*}}{\varepsilon_0^*}\right)$$

$$\frac{E(\varepsilon^* - \varepsilon_0^*)}{\sigma_0^* - \sigma^*} = 0.867\left(\frac{T}{t} - 1\right) \tag{38}$$

Now we define the elastic-follow-up factor $rM = \frac{E(\varepsilon^* - \varepsilon_0^*)}{\sigma_0^* - \sigma^*}$, then:

$$rM = 0.867\left(\frac{T}{t} - 1\right) < \frac{T}{t} - 1 \tag{39}$$

A.3 Third approach: Equivalent stress–strain relation Ramberg-Osgood relation under multiaxial configurations

For multidirection loads, the Ramberg-Osgood relation is generalized as below:

$$\varepsilon^* = \frac{\sigma^*}{E} + \left(\frac{\sigma^*}{K}\right)^{1/n} \tag{40}$$

By converting the tensorial relation of Equation 19 into its equivalent stress–strain form, the Ramberg–Osgood relation can be rewritten as follows:

$$\begin{aligned}\varepsilon^* &= \frac{\varphi\sigma_0^*}{E} + \left(\frac{\varphi\sigma_0^*}{K}\right)^{1/n} \\ &= \varphi\frac{3\varepsilon_0^*}{2(1+\nu)} + \varphi^{1/n}\left(\frac{\sigma_0^*}{K}\right)^{1/n} \\ &= \varphi\frac{3\varepsilon_0^*}{2(1+\nu)} + \phi\varepsilon_0^{p*}\end{aligned} \tag{41}$$

With ε_0^{p*} is the equivalent plastic deformation obtained by Ramberg-Osgood law using the elastic stress σ_0^*.

For the elastoplastic analysis (point **SP**), the local equilibrium equation is given by $\sigma_{ij,j} + f_i = 0$ (with $i, j = \overline{1,3}$). Considering an imposed displacement field $\vec{u}$ prescribed on the boundary of the domain, together with a thermal expansion load $\bar{\bar{\epsilon}}$ acting on all material points (see Figure 18), application of the divergence theorem and the principle of virtual work leads to the following relation involving an infinitesimal variation of the stress field, $\delta\sigma$:

$$\iiint \delta\bar{\bar{\sigma}} : \bar{\bar{\varepsilon}}\, dV = \iint \delta\bar{\bar{\sigma}}\, \vec{n}\, \vec{u}\, dS - \iiint \delta\bar{\bar{\sigma}} : \bar{\bar{\epsilon}}\, dV \tag{42}$$

Using relation (19) and introducing the variation of the unknown field, $\delta\varphi$, yields

$$\iiint_V (\delta\varphi)\, \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}\, dV = \iint_S (\delta\varphi)\, \bar{\bar{\sigma}}_0 \cdot \vec{n} \cdot \vec{u}\, dS - \iiint_V (\delta\varphi)\, \bar{\bar{\sigma}}_0 : \bar{\bar{\epsilon}}\, dV$$

$$\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}\, dV = \iint_S \bar{\bar{\sigma}}_0 \cdot \vec{n} \cdot \vec{u}\, dS - \iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\epsilon}}\, dV \tag{43}$$

For the elastic solution (point **E**), consider the same secondary loads ($\vec{u}$ and $\bar{\bar{\epsilon}}$). The principle of virtual work gives

$$\frac{1}{2}\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}_0\, dV = \frac{1}{2}\iint_S \bar{\bar{\sigma}}_0 \cdot \vec{n} \cdot \vec{u}\, dS - \frac{1}{2}\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\epsilon}}\, dV \tag{44}$$

The energy balances for the elastic solution (Equation (44)) and the elastoplastic solution (Equation (43)) yield

$$\begin{aligned}\iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}\, dV &= \iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}_0\, dV, \\ \iiint_V \frac{1}{\varphi}\, \bar{\bar{\sigma}} : \bar{\bar{\varepsilon}}\, dV &= \iiint_V \bar{\bar{\sigma}}_0 : \bar{\bar{\varepsilon}}_0\, dV\end{aligned}$$

Neglecting the contribution of volumetric strains to the strain energy, the above tensorial relation can be expressed in terms of the equivalent stress and equivalent strain as

$$\iiint_V \frac{1}{\varphi}\, \sigma^* \varepsilon^*\, dV = \iiint_V \sigma_0^* \varepsilon_0^*\, dV$$

Substituting Equations (19) and (41) into the above expression yields

$$\iiint_V \sigma_0^* \left(\varphi\frac{3\varepsilon_0^*}{2(1+\nu)} + \phi\varepsilon_0^{p*}\right) dV = \iiint_V \sigma_0^* \varepsilon_0^*\, dV$$

$$\left(\varphi\frac{3}{2(1+\nu)} - 1\right)\iiint_V \sigma_0^* \varepsilon_0^*\, dV + \phi\iiint_V \sigma_0^* \varepsilon_0^{p*}\, dV = 0$$

Defining the global resilience T as below:

$$\frac{1}{T} = \frac{\iiint_V \sigma_0^* \varepsilon_0^{p*}\, dV}{\iiint_V \sigma_0^* \varepsilon_0^*\, dV}$$

The above equation can be rewritten as

$$\left(\varphi \frac{3}{2(1+\nu)} - 1\right) T + \phi = 0 \tag{45}$$

Now, starting from equation (41), we have:

$$\varepsilon^* = \varphi \frac{3\varepsilon_0^*}{2(1+\nu)} + \phi \varepsilon_0^{p*}$$

$$\varepsilon^* - \varepsilon_0^* = \varepsilon_0^* \left(\varphi \frac{3}{2(1+\nu)} - 1\right) \left(1 + \frac{\phi}{\varphi \frac{3}{2(1+\nu)} - 1} \frac{\varepsilon_0^{p*}}{\varepsilon_0^*}\right)$$

$$\frac{\varepsilon^* - \varepsilon_0^*}{\left(\varphi \frac{3}{2(1+\nu)} - 1\right) \varepsilon_0^*} = 1 + \frac{\phi}{\varphi \frac{3}{2(1+\nu)} - 1} \frac{\varepsilon_0^{p*}}{\varepsilon_0^*}$$

$$\frac{\varepsilon^* - \varepsilon_0^*}{\left(\varphi \frac{3}{2(1+\nu)} - 1\right) \left(\frac{2}{3} \frac{1+\nu}{E} \sigma_0^*\right)} = 1 + \frac{\phi}{\varphi \frac{3}{2(1+\nu)} - 1} \frac{\varepsilon_0^{p*}}{\varepsilon_0^*} \qquad \text{(from equation (49), see Appendix D)}$$

$$\frac{E(\varepsilon^* - \varepsilon_0^*)}{\varphi \sigma_0^* - \frac{2(1+\nu)}{3} \sigma_0^*} = 1 + \frac{\phi}{\varphi \frac{3}{2(1+\nu)} - 1} \frac{\varepsilon_0^{p*}}{\varepsilon_0^*}$$

$$\frac{E(\varepsilon^* - \varepsilon_0^*)}{0.867\sigma_0^* - \sigma^*} = -1 - \frac{\phi}{\varphi \frac{3}{2(1+\nu)} - 1} \frac{\varepsilon_0^{p*}}{\varepsilon_0^*}$$

With the local resilience defined as $t = \frac{\varepsilon_0^*}{\varepsilon_0^{p*}}$ and using equation (45), the above formula becomes:

$$\frac{E(\varepsilon^* - \varepsilon_0^*)}{0.867\sigma_0^* - \sigma^*} = \frac{T}{t} - 1$$

Now we define the elastic-follow-up factor $rM = \frac{E(\varepsilon^* - \varepsilon_0^*)}{\sigma_0^* - \sigma^*}$, then:

$$rM < \frac{E(\varepsilon^* - \varepsilon_0^*)}{0.867\sigma_0^* - \sigma^*}$$

or

$$rM < \frac{T}{t} - 1 \tag{46}$$

B Schematic interpretation of the elastic follow-up curve

This appendix reproduces, for completeness, the mathematical derivation of the elastic follow-up factor r_M from the energy balance of a purely elastic finite-element analysis, following R. Roche's original method [4] as re-derived, in full native LaTeX (no scanned material), from an internal technical note accompanying the source presentation [5]. This is the rigorous justification of Eqs. (9)–(10) used in Section 2.2. Figures 19 and 30 reproduce the graphical and documentary material from which the algebraic steps below were reconstructed.

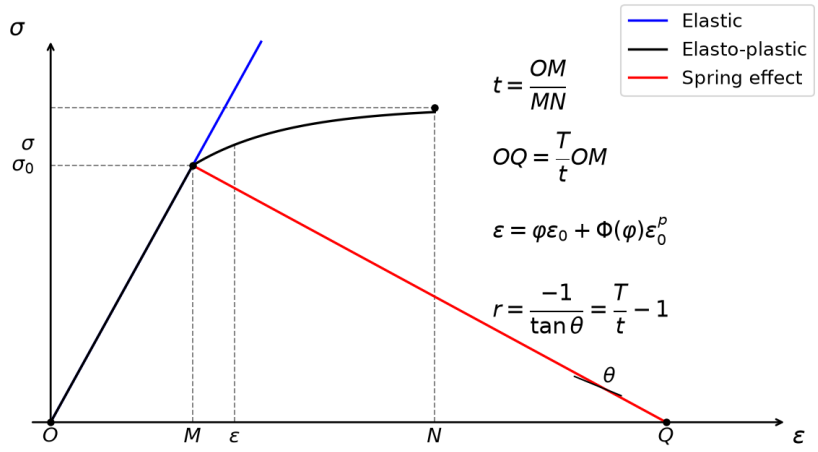


Figure 19: Graphical construction of the elastic follow-up factor r and local resilience t from the elastic (O–M–N) and elasto-plastic (O–ε–Q) response curves at a given imposed strain: $t = OM/MN$ and the elastic-follow-up ("spring effect") line through (ε, σ) and Q has slope $-\tan\theta = -1/(T/t - 1)$.

C Applications of Roche's method

The factor r_M has the following limiting behaviour:

- $r_M = 0$ ($T = t$): no elastic follow-up; the stress is purely secondary (fully self-limiting).
- $r_M \to \infty$ ($T \gg t$): very significant elastic follow-up; the stress behaves as purely primary.

Note that a *higher* r_M is treated by the code as a *more conservative*, more primary-like, assumption; taking $r_M(\text{coeff} = 1) > r_M(\text{coeff} = 0.867)$ is therefore itself a conservative simplification.

In the Roche three-bar analogy, Bar 2 (no spring) corresponds to $r_M = 0$, Bar 1 (rigid connection) corresponds to $r_M \to \infty$, and Bar 3 (elastic spring) lies at an intermediate, finite value of r_M that depends on the spring stiffness k_2.

Operationally, the presentation summarises the RCC-MRx workflow in four steps: (i) perform a purely elastic piping calculation; (ii) deduce the local plastic strain function $\varepsilon_{M,p}(\sigma_{\text{ref}})$ from the adopted Ramberg–Osgood law; (iii) compute the global resilience T by stress-squared weighting of the elastic reference field over the whole structure; and (iv) retain the governing local value of r_M (and therefore of g) at the critical section. Figures 20 and 18 summarise these complementary viewpoints: a graphical construction on the constitutive curve and an energy-balance route derived from the elastic FE solution. In other words, the code is intended to recover an elasto-plastic *estimate* from an elastic base calculation rather than to replace full non-linear analysis in all circumstances.

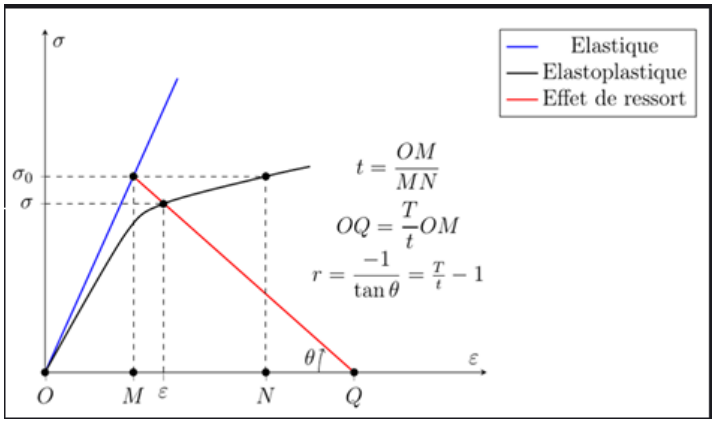


Figure 20: Graphical construction of r_M, reproduced directly from the presentation: $\sigma_0 = \sigma_{\text{el}}$ is the elastic reference stress at point M, σ the actual elasto-plastic stress reached at the same strain ε; the local resilience is $t = OM/MN$, the global resilience T is such that $OQ = (T/t)\,OM$, and the spring-effect line of slope $-1/r_M = \tan\theta$ passes through M and Q.–> to be replaced by a latex code + make it similar to Figure 2

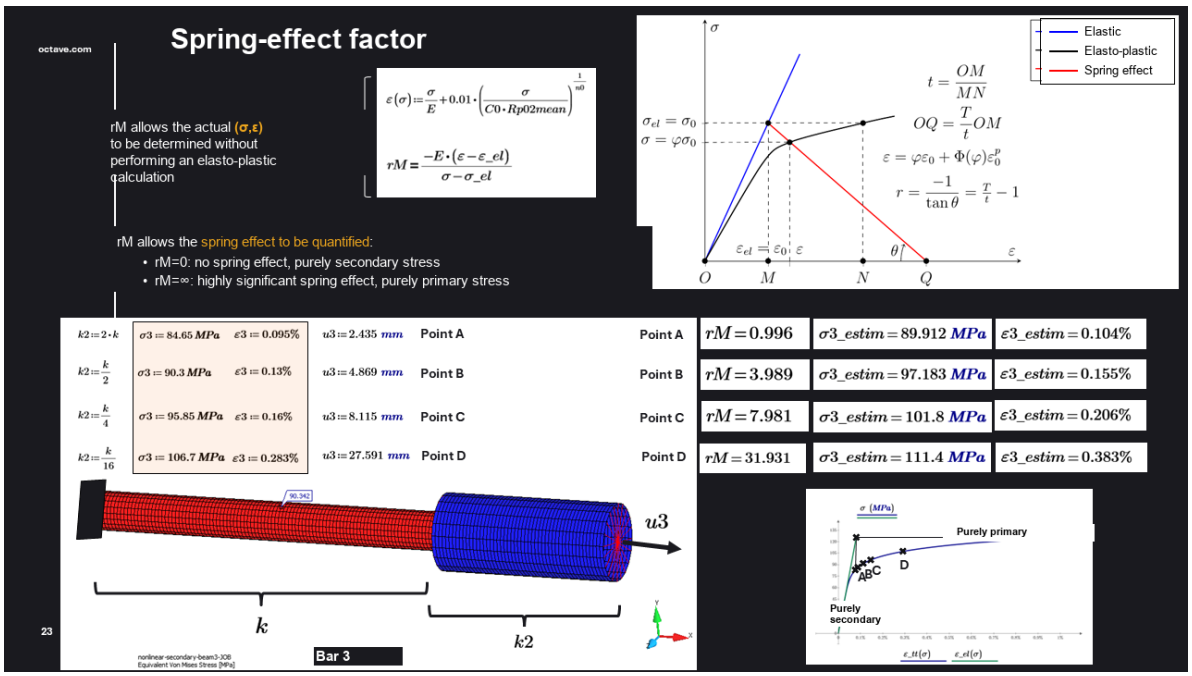


Figure 21: Parametric three-bar sweep extracted from the presentation: reducing the added spring stiffness from $k_2 = 2k$ to $k_2 = k/16$ drives Bar 3 continuously away from the purely secondary limit (Bar 2) and towards the primary-like limit (Bar 1). The slide reports the corresponding sequence of follow-up factors ($r_M = 0.996$, 3.989, 7.981, 31.931), stress estimates ($\sigma_{3,\mathrm{est}} = 89.912\,\mathrm{MPa}$, $97.183\,\mathrm{MPa}$, $101.8\,\mathrm{MPa}$ and $111.4\,\mathrm{MPa}$) and end displacements ($u_3 = 2.435\,\mathrm{mm}$, $4.869\,\mathrm{mm}$, $8.115\,\mathrm{mm}$ and $27.591\,\mathrm{mm}$).–> to redesign. Don't let the rM curve occur too many times

D Definition of equivalent stress and strain

The equivalent stress and equivalent strain for the purely elastic solution are defined as

$$\varepsilon_0^* = \sqrt{\frac{2}{3} e_{0ij} e_{0ij}}, \qquad e_{0ij} = \varepsilon_{0ij} - \frac{1}{3}\varepsilon_{0kk}\delta_{ij}, \tag{47}$$

and

$$\sigma_0^* = \sqrt{\frac{3}{2} s_{0ij} s_{0ij}}, \qquad s_{0ij} = \sigma_{0ij} - \frac{1}{3}\sigma_{0kk}\delta_{ij}. \tag{48}$$

Using Hooke's law $\varepsilon_{0ij} = \frac{1+\nu}{E}\sigma_{0ij} - \frac{\nu}{E}\sigma_{0kk}\delta_{ij}$, and consider the trace of Hook's tensorial relation, we have:

$$\varepsilon_{0kk} = \frac{1+\nu}{E}\sigma_{0kk} - 3\frac{\nu}{E}\sigma_{0kk} = \frac{1-2\nu}{E}\sigma_{0kk}$$

Substituting the trace and tensorial relation of the Hook law into Equation (47) yields

$$e_{0ij} = \frac{1+\nu}{E}\sigma_{0ij} - \frac{\nu}{E}\sigma_{0kk}\delta_{ij} - \frac{1}{3}\frac{1-2\nu}{E}\sigma_{0kk}\delta_{ij}.$$

After simplification,

$$e_{0ij} = \frac{1+\nu}{E}\left(\sigma_{0ij} - \frac{1}{3}\sigma_{0kk}\delta_{ij}\right) = \frac{1+\nu}{E}s_{0ij}.$$

Using the above relation, equations (47) and (48), the equivalent strain is related to the equivalent stress by

$$\varepsilon_0^* = \frac{2}{3}\frac{1+\nu}{E}\sigma_0^*. \tag{49}$$

E FEA 3D model

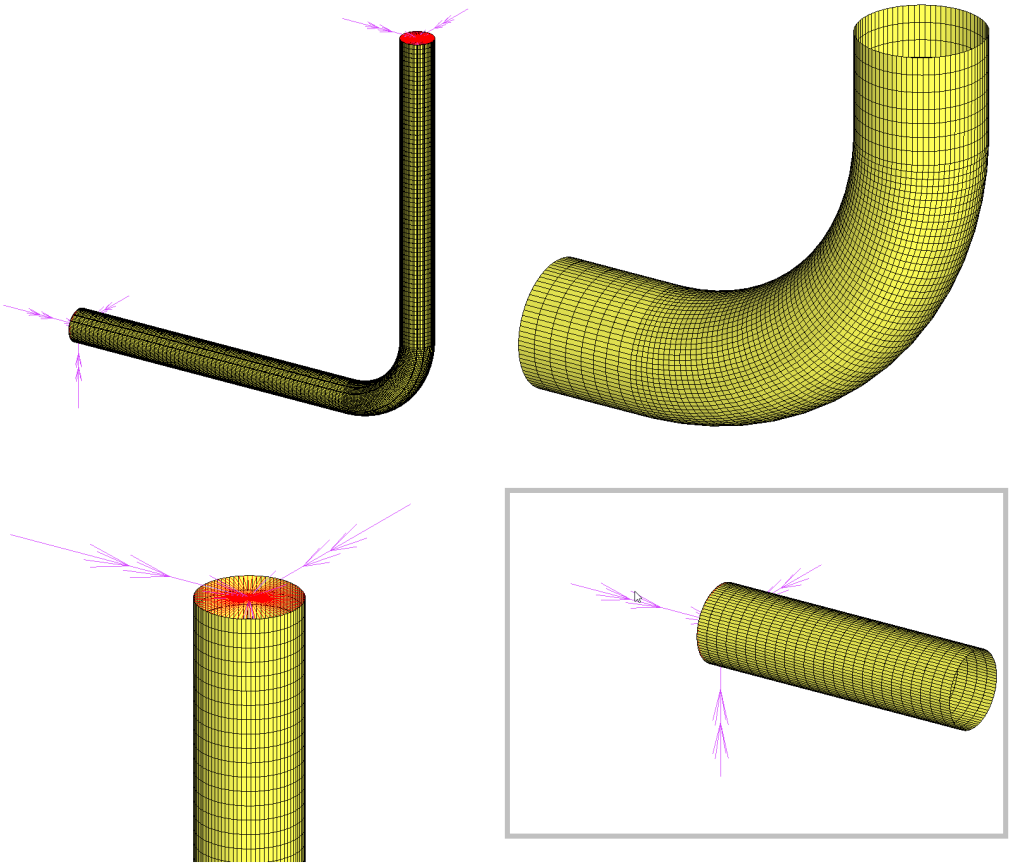


Figure 22: Additional shell-model views extracted from the presentation for the 80-elements-per-circumference discretisation used in the elasto-plastic benchmark: overall and zoomed renderings of the mesh density retained around the elbow, together with the boundary-condition/constraint markers checked before running the non-linear calculation.

F Additional validation: comparison across wall-thickness variants

As a further, independent check, the RCC-MRx abatement coefficient g obtained from the elasto-plastic FE reference (g_{FEA}) was compared against the RCC-MRx analytical value ($g_{\text{RCC-MRx}}$) on two additional model variants of the same benchmark, differing only in wall thickness (12 mm and 15 mm), including large-strain and plasticity effects. The comparison is summarised in Figure 23. In both cases, the RCC-MRx value bounds the FE-derived value, confirming the conservatism of the code-mandated coefficient across a range of section stiffnesses.

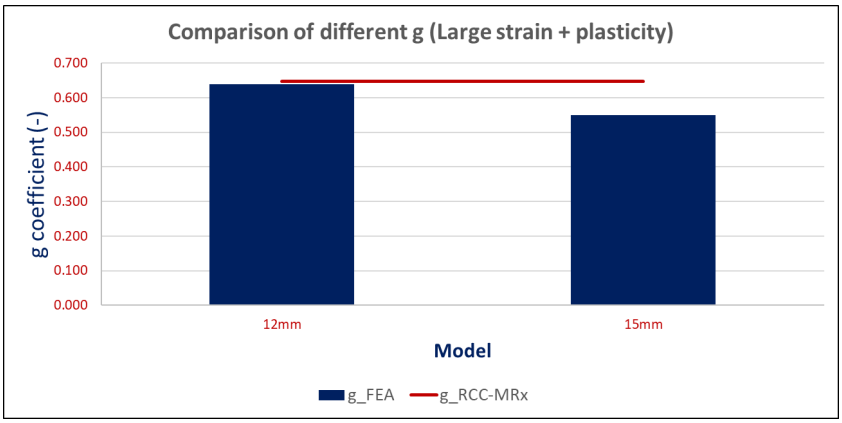


Figure 23: Comparison of the abatement coefficient g from a large-strain elasto-plastic FE calculation (g_{FEA}) against the RCC-MRx analytical value ($g_{\text{RCC-MRx}}$), for two wall-thickness variants (12 mm and 15 mm) of the benchmark elbow. The RCC-MRx value remains conservative (equal to or above g_{FEA}) in both cases.

F.1 Analytical validation on a cantilever with imposed end displacement

To validate the production-code implementation of r_M and g independently of the elbow benchmark, a simple, closed-form case is considered: a straight cantilever beam of length l, clamped at one end, subjected to an imposed transverse displacement Δu at the free end (Figure 24). In the benchmark slide deck, the imposed displacement is specified explicitly as $\Delta u = 12\,\mathrm{mm}$. In the closed-form derivation below, the curvilinear abscissa x is measured from the free end ($x = 0$) towards the clamp ($x = l$), so that the elastic bending moment and the elastic reference stress vary linearly with x.

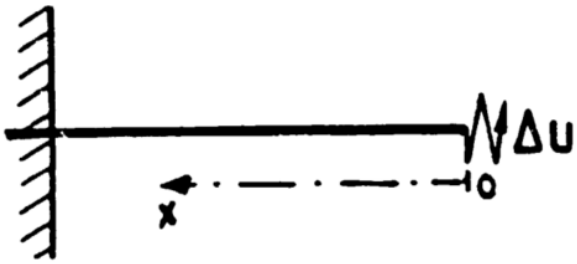


Figure 24: Schematic of the cantilever validation case: a beam of length l, clamped at one end and subjected to an imposed transverse displacement Δu at the other; for the analytical derivation, x is measured from the free end to the clamped end.

For this configuration, the elastic reference stress is linear in x, $\sigma_{\mathrm{ref}}(x) = \omega\, x$ with $\omega = 0.79\,(3EI/l^3)\,\Delta u/Z$, and the local resilience $1/t(x)$ follows directly from the Ramberg–Osgood plastic strain $\varepsilon_{M,p}$. Carrying out the integrals of Eq. (10) in closed form for this linear stress distribution gives the elastic follow-up factor as an explicit function of position,

$$r(x) \;=\; \frac{T}{t} - 1 \;=\; \frac{1 + 2n_0}{3n_0} \left(\frac{x}{l}\right)^{\frac{1-n_0}{n_0}} - 1, \tag{50}$$

which is monotonically increasing towards the clamp and is therefore maximised at the clamped end ($x = l$), giving the closed-form bound

$$r_{\max} \;=\; \frac{1 - n_0}{3n_0}. \tag{51}$$

For the benchmark material ($n_0 = 0.1125$), the same closed-form development gives the global resilience quantity $T = 0.444$, and Eqs. (50)–(51) then give $r_{\max} = 2.63$. Using the no-pressure RCC-MRx expression for the abatement coefficient on that analytical solution yields $g = 0.3118$, i.e. essentially the same value as the production-code result $g = 0.3116$ reported below. Figures 25, 26 and 27 show, respectively, the FE model, the non-linear stress field and the production-code output used to validate this closed-form result.

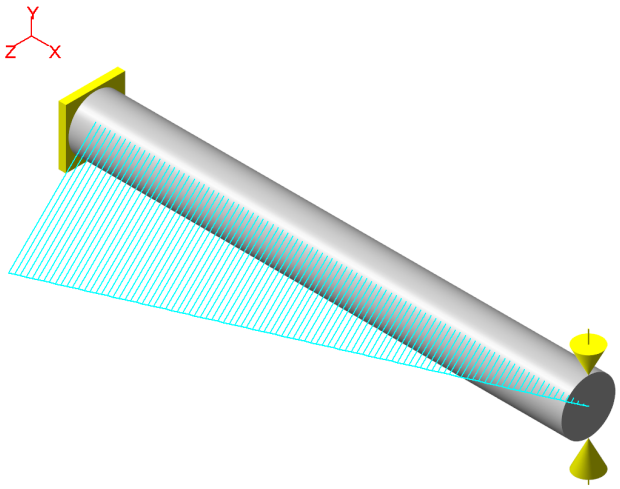


Figure 25: Finite-element model of the cantilever validation case: clamped at one end (yellow anchor), imposed displacement enforced through a rigid link at the free end (yellow cones).

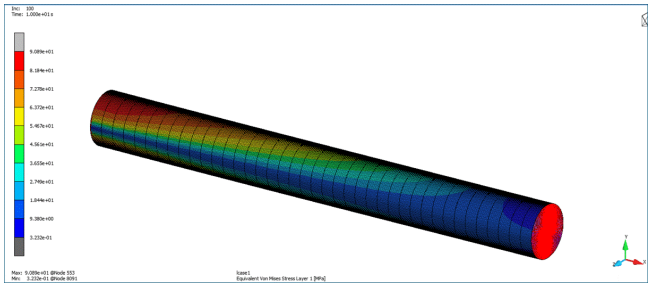


Figure 26: Elasto-plastic MSC Marc result for the simplified validation benchmark used to check the r_M/g implementation: the equivalent von Mises field confirms the stress localisation pattern that is then compared against the elastic-plus-RCC-MRx route.

□ H E X A G O N K-7.1 PAGE NO. 17

++ HEXAGON/NEW SOLVER ++ RCC-MRx type P Version 01.00.00.0295 PC-ISL-64b Release: Jul 2025

The following table summarizes the calculations of monotonic elastic follow-up (max value rM) and abatement coefficient g:

	LOAD CASE NO. 501											
Element	SigmaRef1 (MPa)	SigmaRef2 (MPa)	EpsMpRef1 (%)	EpsMpRef2 (%)	Max(1/t)	Sigma (MPa)	SigmaP (MPa)	EpsMpSig (%)	EpsMpSigP (%)	T	rM	g
0	118.04	116.86	0.523	0.478	6.86	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
1	116.86	115.68	0.478	0.437	6.34	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
2	115.68	114.50	0.437	0.399	5.85	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
3	114.50	113.32	0.399	0.363	5.40	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
4	113.32	112.14	0.363	0.331	4.97	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116

Figure 27: Production-code output (Aspect Nuclear Pipe Stress / HEXAGON NEW SOLVER, RCC-MRx type P) for the cantilever validation case (LC 501): the code reports $r_M = 2.627$ at the critical section, in close agreement with the closed-form bound $r_{\max} = 2.63$ of Eq. (51), together with $g = 0.3116$.

The close agreement between the closed-form analytical bound $r_{\max} = 2.63$ and the production-code result ($r_M = 2.627$) constitutes an independent, purely analytical validation of the code's implementation of the RCC-MRx elastic follow-up factor and abatement coefficient, on a case entirely decoupled from the elbow ovalisation and shell-modelling effects discussed above.

G A simple example of Roche method

$$L := 100\ in = 2.54\ m \qquad E := 192000\ MPa$$

$$L_faible := 32\ in = 0.813\ m \qquad K := 793\ MPa$$

$$L_forte := L - L_faible = 1.727\ m \qquad m := \frac{1}{2.915}$$

$$\Delta\varepsilon(\Delta\sigma) := \frac{\Delta\sigma}{E} + 0.01 \cdot \left(\frac{\Delta\sigma}{K}\right)^{\frac{1}{m}}$$

Figure 28: Numerical data from Roland Roche's original worked example (straight bar of total length $L = 2.54\,\mathrm{m}$, split into a flexible segment $L_{\mathrm{weak}} = 0.813\,\mathrm{m}$ and a stiff segment $L_{\mathrm{stiff}} = 1.727\,\mathrm{m}$, with $E = 192\,000\,\mathrm{MPa}$, Ramberg–Osgood parameters $K = 793\,\mathrm{MPa}$, $m = 1/2.915$), illustrating the calculation of the elastic follow-up factor on a simple two-segment bar.

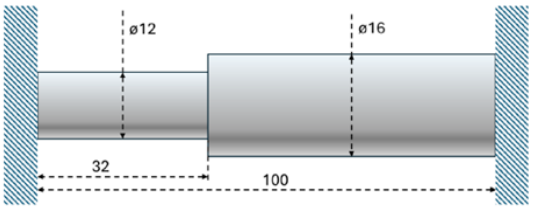


Figure 29: Geometry of Roland Roche's original two-segment bar example: a stepped shaft with a flexible segment of diameter 12 mm and length 32 in and a stiffer segment of diameter 16 mm, clamped at both ends, with the total length (100 in) measured from the flexible end.

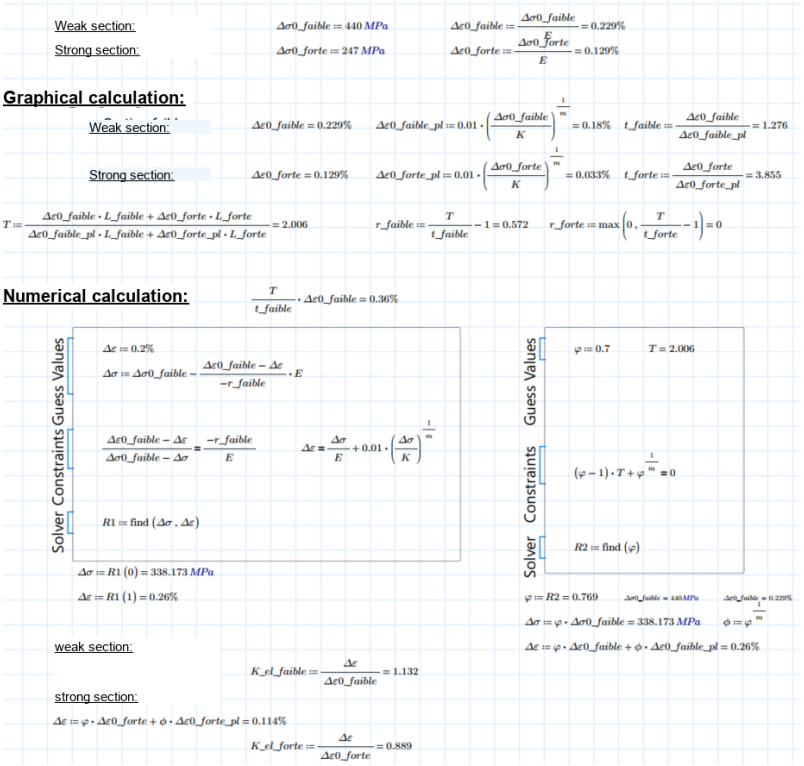


Figure 30: Numerical worksheet from the source presentation reproducing Roche's two-section example (geometry in Figure 29): the elastic reference stresses ($\Delta\sigma_{0,\mathrm{weak}} = 440\,\mathrm{MPa}$, $\Delta\sigma_{0,\mathrm{strong}} = 247\,\mathrm{MPa}$) give local resiliences $t_{\mathrm{weak}} = 1.276$ and $t_{\mathrm{strong}} = 3.855$, combined into a global resilience $T = 2.006$ and local elastic follow-up factors $r_{\mathrm{weak}} = T/t_{\mathrm{weak}} - 1 = 0.572$ and $r_{\mathrm{strong}} = \max(0, T/t_{\mathrm{strong}} - 1) = 0$; the lower numerical-calculation block solves the same problem with Mathcad solve blocks, confirming $\varphi = 0.769$ and $\Delta\sigma = 338.173\,\mathrm{MPa}$ by an independent route.

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